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Conditions for the Use of Wh-Clefts

Corpus-Based Analysis of Wh-Clefts in TED-Talks

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1 Introduction

In TED Talks, speakers address a wide range of topics – from the climate crisis to technical innovations, scientific breakthroughs, political issues, and beyond (TED Conferences 2025). In these talks, it is not only important what information is conveyed, but also how this information is conveyed in order to reach the audience in the best possible way. To understand and recognize which information is particularly important, it is crucial that information within sentences is structured in a meaningful way and linked logically to the discourse. One strategy to achieve this is using wh-cleft sentences. With their two-part structure, wh-clefts allow speakers to adress familiar aspects of the content first and then introduce and highlight new information (Prince 1978). By structuring the information in a linear way and shifting the focus to the new information, wh-clefts serve as an important devise to meet the discourse needs of a successful talk.

Previous research, particularly by Prince (1978) and Gast/Wiechmann (2012) argued that wh-clefts typically follow this specific information structure pattern. In this structure the wh-clause picks up information that is accessible for the hearer and thus given or inferable from the context. The subsequent post-copular constituent then introduces information that is new to the hearer or at least less familiar than the information contained in the wh-clause. From this point of view, wh-clefts are useful tools to achieve a coherent discourse and a logical flow of information.

Since many articles focus either on comparing it-clefts and wh-clefts or on cross-linguistic comparisons of wh-clefts (e.g., Prince 1978, Declerck 1984, Hedberg 2000, Lambrecht 2001, Collins 2006, Gast/Wiechmann 2012), but only a few articles deal exclusively with wh-clefts and their specific conditions in spoken discourse, it is worth examining this matter in more detail. Accordingly, this paper examines the conditions for the use of wh-clefts in spoken discourse, by using TED Talks as a data source. It will be explored whether wh-clefts in fact consistently follow the typical information structure, in which the familiar information is presented in the wh-clause and newer information subsequently in the post-copular constituent. In addition, it will be analyzed how wh-clefts function in relation to the preceding discourse and what kind of anchor information they typically refer to. Finally, the weight distribution within wh-clefts is examined in more detail and it

is investigated whether the hypothesis that the post-copular constituent is always heavier than the wh-clause and whether the weight of the constituents is connected to their information status is true. For this analysis, a corpus study was conducted in which a total of 100 wh-clefts from the TED corpus were assessed.

The structure of the paper is as follows. First, the theoretical foundations of wh-clefts will be covered, followed by explaining the methodological approach of the corpus analysis. In the final part the results of the corpus study will be presented and interpreted in more detail.

2 Theoretical Background

In general, a cleft construction can be defined as a complex sentence structure consisting of a relative or at least relative-like clause and a main clause including a copula (Lambrecht 2001: 467). There are two different types of cleft sentences in the English language – *it-clefts* and *wh-clefts*. Since this paper analyzes the conditions for the use of wh-clefts, the focus will be solely on wh-clefts, also known as *pseudo-clefts*. The terminology that is used in this paper is wh-clefts. In the following a simple example illustrates how wh-clefts differ from non-clefted sentences.

1) a. Lisa found her dog.
   b. What Lisa found was her dog.

Example 1a shows a non-clefted sentence with a canonical SVO (subject – verb – object) word order. Sentence 1b shows an example for a wh-cleft in a non-canonical word order. From a cognitive perspective, these two sentences are considered synonymous because they contain the same information (Prince 1978: 883). However, these sentences cannot be used interchangeably as they fulfill different functions in terms of presupposition (Prince 1978: 884). A possible presupposition for the example could be that *Lisa found something*. If a sentence truly serves as a presupposition for another sentence this can be verified by negating the sentence in question (Prince 1978: 884).

2) a. Lisa didn't find her dog.

b. What Lisa found wasn't her dog.

In sentence 2a it could be that Lisa did not find anything at all, which would mean that the presupposition *Lisa found something* is not true. In the clefted version, however, the statement that Lisa found something remains true, even after negating the sentence. This shows that wh-clefts and non-clefted sentences differ in terms of presupposition (Prince 1978: 884).

2.1 Structure of wh-clefts

The formation of a wh-cleft sentence leads to a syntactic division of originally one clause into two distinct clauses (Gast/Wiechmann 2012: 340). Therefore, wh-cleft sentences consist of two main sentence elements – the wh-clause (e.g. *what we need* …) and the copular predicate (see Figure 1). The copular predicate is composed of a copular verb and its complement (e.g. … *is decentralization*). In the following the term *post-copular constituent* will be used to refer to the element occuring after the copula.

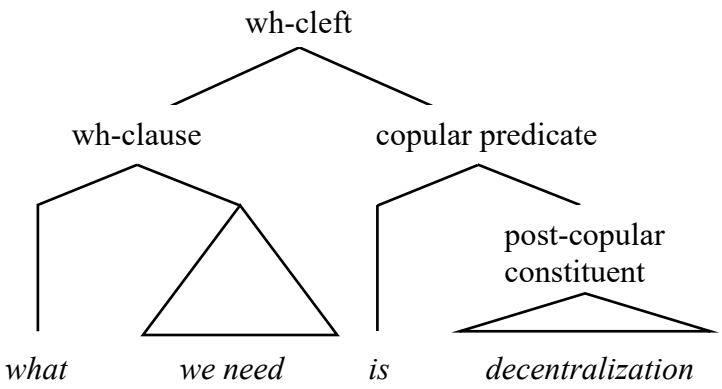

***Figure 1*:** Structure of a wh-cleft

(adapted from Gast/Wiechmann 2012: 340).

Usually, the information contained in the post-copular constituent forms the highlighted part of the sentence and thus holds the information on which the focus is placed. *What* in the wh-clause then refers to this highlighted sentence element

(Herriman 2004: 447). From a semantic point of view there is an identity relation between the wh-clause and the highlighted post-copular constituent. In the above example, the wh-clause *what we need* serves as an identifier for the highlighted part, in this case *decentralization* (Herriman 2004: 448). Declerck further specified this relationship, by describing the wh-clause as a variable that needs to be satisfied by the value, which is the post-copular constituent. This means that the value *decentralization* is assigned to the variable *what we need* (1984: 252). The semantic function of wh-clefts is thus to determine the value that complies with the definition in the variable (Herriman 2004: 448).

Pragmatically speaking, a wh-cleft can be divided into a question (*questio*) and answer (*responsio*). The wh-clause contains information that corresponds to a question that the hearer is assumed to have in mind at the time of the utterance (Gast/Wiechmann 2012: 338). Klein/von Stutterheim (1987,1992) introduced the *Questio-Modell* to describe the discourse conditions for wh-clefts. According to this model, the wh-clause expresses the question currently under discussion, meaning the question present in the hearer's mind at the moment of hearing the utterance. The post-copular constituent then contains the corresponding answer to this question. Wh-clefts can therefore be regarded as question-answer pairs within a single sentence (Gast/Wiechmann 2012: 339). Based on the previous example *What we need is decentralization,* the following *questio* and *responsio* would be generated.

3) questio: what do we need?
4) responsio: decentralization

From this it follows that the discourse conditions for wh-clefts are only fulfilled if the clause expressed in the wh-clause corresponds to a question that is part of the context and can therefore be assumed to be present in the hearer's mind (Gast/Wiechmann 2012: 339).

2.2 Types of Wh-Clefts

The five different types of cleft constituents that are possible in English are: noun phrases (NP-clefts), complementizer phrases (CP-clefts), verb phrases (VP-clefts),

to-infinitive phrases (TP-clefts) and gerund phrases (GP-clefts) (Gast/Wiechmann 2012: 343). The following section provides examples from the TED Corpus to illustrate each of the types of wh-clefts.

5) NP-Cleft: "What they had in common was [$_{NP}$ a sense of courage]. (TED-ID15)

6) CP-Cleft: "What is happening is [$_{CP}$ that, very rapidly in historical terms, the world is being driven and shaped, not by the old developed countries, but by the developing world]". (TED-ID17)

7) VP-Cleft: "What we did was [$_{VP}$ expose American babies during this period to Mandarin". (TED-ID25)

8) TP-Cleft: "What we really yearn for as human beings is [$_{TP}$ to be visible to each other]". (TED-ID28)

9) GP-Cleft: "What that means is [$_{GP}$ prepping my outfit, (Laughter) prepping options, trying to figure out what I 'm coming behind and going in front of]". (TED-ID19)

2.3 Motivations for Wh-Clefts

There are several motivations for the use of wh-clefts. First, wh-clefts have the function to structure a sentence linearly. By dividing the sentence into *questio* and *responsio,* the information structure and syntax are presented in a linear order. In a regular, non-clefted sentence, elements of *questio* and *responsio* would be mixed and thus more difficult to understand (Gast/Wiechmann 2012: 351).

In addition to the linear structuring and structural separation of *questio* and *responsio*, there are also semantic reasons for using wh-clefts. It is common that the information contained in the wh-clause fulfills a commenting function and does not contribute to the information of the actual proposition. Wh-clefts thus help to separate commenting expressions, from the actual proposition and thus to distinguish less important from more important information. Since the wh-clause

and the cleft-constituent are located on different levels – the propositional and the illocutionary level – this phenomenon is referred to as 'level separation' (*Ebenentrennung*) (Gast/Wiechmann 2012: 356).

A syntactically motivated function of wh-clefts is that it places the heavier constituent further to end of the sentence. According to Behagel's 'law of growing elements' (*Gesetz der wachsenden Glieder*) the heavier constituent should always follow the lighter one (Behagel 1930 in Arnold 2000: 29). Quirk et al. (1972) referred to the same rule as principle of 'end-weight' (Arnold 2000: 29). It is advantageous for both the speaker and the hearer if the more complex constituent is placed at the end, thereby achieving a right-heavy structure. For the speaker it is helpful, as it makes it easier to plan the utterance and allows more time to formulate longer and more complex sentence constituents. It is also helpful for the hearer as it makes it easier to process the information in a sentence if the heavier and often also more complex constituent comes last (Arnold 2000: 32)

There are different opinions on how heaviness can be measured. In the original notion by Behagel, heaviness depends only on the number of words that a constituent consists of and thus heaviness could be measured by counting the words of each consituent. Other linguists, such as Chomsky, define heaviness not only in terms of word count, but also in terms of complexity (Arnold 2000: 29).

Another important motivation for the use of wh-clefts is to structure a sentence in terms of given and new information. The following chapter explains the notion of givenness with regard to wh-clefts in more detail.

2.4 Information Structure of Wh-Clefts

With regard to the analysis of the information status of wh-clefts an important notion was introduced by Wallace L. Chafe in 1974. He stated that each sentence consists of both given and new 'material'. In his binary distinction, Chafe defines information as given if it refers to information "which the speaker assumes is already in the addressee's consciousness …" (1974: 112) and new information which is assumed to be not in the hearer´s consciousness at the moment of hearing an utterance (Chafe 1974: 112). The binary distinction between given and new introduced by Chafe, was further elaborated by the taxonomy of givenness provided by Prince. She made a distinction between new, inferable and evoked information.

Evoked information refers to information that is either given textually or situationally (Prince 1981 in Loock 2013: 70). Information that has not been mentioned or could not be inferred, counts as new information (Prince 1992 in Ward & Birner 2008: 129). Prince already noted that the information delivered in the post-copular constituent does not have to be entirely new, but at least it must be less accessible to the hearer or reader than the information in the wh-clause (Prince 1978: 892). Important to add is that the information contained in the wh-clause does not actually have to be in the hearer's mind but simply must be appropriate and therefore cooperatively assumable to be present in the hearer´s mind (Prince 1978: 889). Therefore, Prince modified Chafe's definition slightly from "assumes" to "may cooperatively assume" (Prince 1987: 894). Accordingly, Prince formulated the discourse condition:

> A WH-cleft will not occur coherently in a discourse if the material inside the (subject) WH-clause does not represent material which the cooperative speaker can assume to be appropriately in the hearer's consciousness at the time of hearing the utterance. (Prince 1978: 888).

Following this condition, it would imply that the wh-clause in a wh-cleft must always contain information that is somehow accessible from the preceding context, because otherwise there would be no possibility that the hearer has the material of the wh-clause currently in mind. Which information then could be considered as given is largely dependent on the linguistic, as well as non-linguistic context or the speech situation itself (Prince 1987: 904). This condition would also imply that wh-clauses usually do not occur at the beginning of a discourse as there is no previous discourse from which the information of the wh-clause could be inferred. Prince provided an example of a wh-cleft serving as a discourse opener (see 10).

10) ## *Hi! What my name is is Ellen. (Prince 1978: 888)

She argued that although it can be inferred from common knowledge that human beings usually have names, it cannot be assumed that the hearer has this information or question in mind at the moment of hearing the utterance. However, there are also exceptions to this rule, as for example a scientific article possibly can begin with something like *What we aim is to show is*…. In this case, a wh-cleft would be

appropriate, because it can be assumed that someone who starts reading a scientific article will wonder what the aim of that article is (Prince 1978: 888).

The knowledge referred to in a wh-cleft and which is presumably present in the hearer's consciousness is called the antecedent (Clark & Haviland 1977: 4). There are different types of antecedents that can serve as anchor information, on the basis of which the information in the wh-clause can be regarded as either given or inferable. In some cases, the information contained in the wh-clause is actually mentioned explicitly in the preceding linguistic context, which means that there is an explicit antecedent. Example 11 shows that the author already explicitly states that something is being said, and this information is more or less repeated in the wh-clause.

11) "…in Haviland and Clark (press) they say that … Certainly WHAT THEY ARE TALKING ABOUT IS …" (Chafe 1976:42 in Prince 1978: 889).

However, since a repetition of explicitly mentioned information is rare, there is another type of antecedent which is implicit information. The two categories, explicit and implicit antecedents, are not always clear-cut and thus sometimes difficult to distinguish. In terms of implicit antecedents, it would be the case that in the preceding linguistic context of a wh-cleft a certain information is given from which the information in the wh-clause can be implicitly derived. This inference takes place via bridge building. Considering example (12) the only inference, or bridge how Haviland & Clark (1974) call it, that needs to be drawn is the realization that if there is any kind of problem, one usually worries about this problem.

12) '"Himself a religious Jew, Prof. Flusser says that Carter's piety is not the problem. "WHAT I'M WORRIED ABOUT", he declares, "is that the influence of Carter's advisers may eclipse the influence of the Divine Spirit"' (JE, 9/10/76: 33 in Prince 1978: 887).

In this example the *problem* can be viewed as the anchor information from which it is inferable that he worries about something. Moreover, antecedents can also have a contrastive relation with the information in the wh-clause. This contrast could possibly be positive/negative, comparative/superlative or antonymous in nature.

Besides the linguistic antecedents, there are also different types of metalinguistic antecedents. This type of antecedent implies that the information in the wh-clause is not explicitly or implicitly given in the preceding linguistic context, but must be inferred from metalinguistic notions (Prince 1978: 890). For example, it can be assumed that when a speaker makes an utterance, he/she is thinking about something and ultimately mean something with this utterance. Considering these metalinguistic notions, wh-clauses like *what I mean…* or *what I think…* can occur and are very common in conversational discourse (Prince 1978: 892).

Another type of antecedent that is also not inferable from the preceding linguistic context but rather from the speech situation is that something is happening. It can be assumed events are constantly taking place and for this reason it can also be assumed that the question of what is happening at the moment would almost always be appropriate in the hearer's mind. Both *think*- and *happen*-clefts often serve to introduce main statements and therefore often do not have a *that*-complementizer (Prince 1978: 893).

2.5 Types of Discourse Relation

For the analysis of discourse reactions, the classification by the Penn Discourse TreeBank 2.0 provides a good basis for this corpus analysis and is therefore briefly described in the following. This classification distinguishes four different classes of semantic relations which are suitable to analyse the relationship between wh-clefts and the preceding context. The four categories of semantic relations are: temporal, contingency, comparison and expansion. A *temporal* semantic relation means that the discourse elements are related in terms of time, either preceding or succeeding one another or indicating an asynchronous event. The category of *contingency* includes semantic relations such as causes (reason, result), conditions (hypothetical, general, unreal present, unreal past, factual present, factual past), as well as pragmatic conditions (relevance, implicit assertion). The category *comparison* applies when the wh-cleft introduces a contrastive idea, for example it expresses either a juxtaposition or opposition. Also, pragmatic contrasts, concessions (expectation, contra-expectation) and pragmatic concessions fall in this class of semantic relations and help to highlight differences or similarities between discourse segments. Finally, the category *expansion* would refer to wh-clefts that

add information or further elaborate on the previous context. This includes conjunctions, instantiations, restatements (specification, equivalence, generalization), alternatives (conjunctive, disjunctive, chosen alternative) exceptions and lists (Prasad et al. 2008: 2965).

3 Corpus Analysis – TED Corpus

This corpus analysis is based on data from the TED Corpus. TED Talks, short for technology-, entertainment- and design talks, are short talks on topics from various fields like psychology, leadership, education, AI, mental health, business, motivation, communication, personal growth, sports, health, language and more. TED's mission is to "discover and spread ideas that spark conversation, deepen understanding, and drive meaningful change" (TED Conferences 2025). Examples from TED Talks are suitable and interesting for investigating the conditions of the use of wh-clefts, as they are spoken language which is more authentic than written literature. However, TED Talks are scripted to a certain extent which makes them not entirely authentic. This could also be seen as an advantage, however, as the fact that these talks are formulated in advance and usually well-planned means that the information conveyed in TED Talks is likely to be richer than in casual conversation. Furthermore, it is advantageous that TED Talks cover many different areas and speech styles, so that the analysis and interpretation of the use of wh-clefts is not limited to a single register. In the following the methodological approach of the corpus analysis will be explained, followed by presenting and interpretating the findings of the analysis.

3.1 Methodology

The first step taken to identify examples of wh-clefts in the TED corpus was to formulate a query that extracts these text passages. The following query (13) was formulated for this purpose with the intention of filtering out classic examples of wh-clefts in TED Talks and minimizing the number of results that contain wh-clauses but are not real wh-clefts.

13) [word="What*"][word="[a-zA-Z0-9]*"]+
[word="is|was|isn't|wasn't||can|can't|could|couldn't|might|might|not|may|
may|not|should|shouldn't|must|mustn't|has|hasn't|been"]

Even though not all wh-clefts begin with *what*, it emerged from the literature on wh-clefts that almost all examples are *what*-clefts and that other types seem to play a rather minor role. For this reason, I have decided to restrict my analysis to what-clefts. The following procedure was then carried out. First, the query was used to search for wh-clefts in the TED Corpus, with the filters set to generate 500 results, displaying 5 sentences before and 5 sentences after each wh-cleft. This is particularly important, as otherwise it would not be possible to determine whether the information presented in the wh-clause is new, given or can be inferred from the preceding context. In retrospect, however, the five sentences after the wh-cleft could have been omitted, as it is primarily the preceding context that matters for the analysis.

Starting with the first search result, each TED Talk excerpt was examined one by one and identified as either a wh-cleft or as no wh-cleft. Every text excerpt, including the examples that were identified as no wh-clefts, were then copied into a separate Word document. Next, all examples containing a wh-cleft were assigned an ID number and the examples containing no real wh-cleft were assigned only a number and were marked in red. This was repeated until 100 examples with wh-clefts had been identified, copied, and assigned ID numbers from 1 to 100.

However, during the analysis, I noticed that 7 of the corpus examples appeared twice and one example had been incorrectly coded as a wh-cleft. For this reason, 8 additional wh-clefts were subsequently included in the analysis. As a result, a total of 108 IDs were assigned, but the number of analyzed wh-clefts remains at 100. The number of 100 wh-clefts was set in order to facilitate the calculation of percentages in the results later. Since not all of the first 100 search results were wh-clefts and considering that several examples appeared twice, a total of 139 TED talk excerpts had to be transferred in order to obtain 100 proper examples for this analysis. In total, 31 text excerpts did not contain a real wh-cleft. Frequent results that were mistakenly presented as wh-clefts in the corpus search included, for example hypothetical questions like *What if you could…?* (TED-1) or wh-interrogative questions like *What is the latest possible moment …?* (TED-6).

Given that the number of search results with no wh-cleft is relatively high, it can be said that the search query is not entirely ideal and could be further optimized.

The aim of this corpus analysis is to investigate whether the examples from the TED corpus confirm the hypothesis that the wh-clause contains given or inferable information and the post-copular constituent introduces new information and whether this information status correlates with the heaviness, cleft-type or the way the wh-cleft is related with the discourse. To examine this, six categories were selected: wh-clause information status, post-copular constituent information status, type of antecedent, discourse relation, cleft type and heaviness. In the following, it will be explained in more detail how the TED Talk examples are classified within these categories and assigned specific codes.

The information status of both the wh-clause and the post-copular constituent is analyzed based on the three-part classification provided by Prince (1981), with the term "evoked information" being slightly modified to given information. Cases in which the information of the wh-clause is explicitly, or at least almost explicitly mentioned in the preceding linguistic context or is present in the speech situation were coded as GIVEN. Examples in which the information provided in the wh-clause could possibly be inferred from preceding context or the speech situation, were coded as INFERABLE. Finally, examples that where neither explicitly mentioned nor could be inferred were coded as NEW. The wh-clefts that could not be clearly assigned to one of these categories were coded as AMBIGUOUS.

The TED Talk examples are also analyzed in terms of the type of antecedent and the relation with the preceding discourse. Following Prince's (1978) classification, the antecedents were coded as LINGUISTIC or METALINGUISTIC. The linguistic antecedents were further differentiated into explicit (EXP), implicit (IMP), and contrastive antecedents (CON). Given that the wh-clefts in question are excerpts from TED Talks and thus spoken language that is sometimes combined with visual elements, an additional category was added and coded as SITUATIONAL. Furthermore, the analysis of the discourse relation is based on the four categories introduced in the Penn Discourse TreeBank 2.0 – TEMPORAL, COMPARISON, CONTINGENCY, EXPANSION. As there were some examples without a real connection to the previous discourse, one more category was added which was labeled and coded as INITIAL.

Finally, in order to measure how heavy the constituents of the wh-cleft are, Behagel's basic notion of heaviness was used, which refers to relative length of the constituents in terms of the number of words. In a study by Arnold et al., which examined newness and heaviness in terms of noun phrases and dative alternation, they also based their measurement of heaviness on this definition (2000: 35). On this basis that they established three categories of heaviness for dative alternation. I adapted these categories for my analysis to measure how the weight is distributed in wh-clefts (see Table 1). The wh-clause was coded as WH and the post-copular constituent as PCC.

1)	PCC – WH = -2 or less	PCC < WH
2)	PCC – WH = between -1 and 1	PCC = WH
3)	PCC – WH = 2 or more	PCC > WH

***Table 1*:** Categories of heaviness for wh-clause and post-copular constituent (adapted from Arnold et al. 2000: 36).

To determine whether the wh-clause or the post-copular constituent is heavier, the words of each constituent was counted (see 14).

14) What I really think we gain from this is students getting intuition and
1 2 3 4 5 6 7 8 1 2 3 4
experience in far greater quantities than they 've ever got before. (TED-ID9)
5 6 7 8 9 10 11 12 13 14 15

In this example from the TED corpus, the wh-clause comprises a total of 8 words and the post-copular constituent a total of 15 words. When the word count of the wh-clause is now subtracted from the word count of the PCC, the result would be 7. This indicates that the post-copular constituent is significantly heavier and therefore the code PCC could be assigned for this example.

3.2 Results & Interpretation

In the following subchapters the results of the corpus analysis for each category will be presented and interpreted and when appropriate correlated with each other.

3.2.1 Information Status

The first category that was analyzed in the TED corpus was the information status of the wh-clause, in terms of given, inferable or new. The results illustrate that in more than half of the TED Talk examples (61%) the information in the wh-clause was inferable either from the previous linguistic context, metalinguistic notions or the speech situation (see Table 2). In 35% of the cases the information of the wh-clause was explicitly mentioned in the previous discourse or explicitly present in the current speech situation and thus these cases were considered as given information. The two categories given and inferable together thus make up 95% of the cases and only in 4% of the cases the information that is provided in the wh-clause was new to the audience.

15) What we have found working with the World Bank … (TED-ID83)

This would be an example in which the information provided in the wh-clause is not entirely given or inferable for the audience, because that they worked with the World Bank was not mentioned in the previous discourse. However, what needs to be taken into consideration is, that in these corpus examples only 5 preceding sentences are provided and therefor it could be the case that the World Bank was somehow mentioned in the previous discourse. Nevertheless, from this text excerpt alone, this information is not given and thus coded as NEW.

GIVEN	INFERABLE	NEW
35	61	4
35%	61%	4%

Table 2: Wh-clause information status corpus analysis results.

In accordance with the hypothesis, the information status of the post-copular constituent was in most of the cases new (90%) (see Table 3). Only in 3 examples (3%) the post-copular constituent provided information that was inferable from the context and in 5 examples (5%) the information was either explicitly mentioned or explicitly visible in the current speech situation and thus coded as GIVEN. One of the examples in which this was the case is the following.

16) He meant that you can see the sky and that you can experience *the sun*. What it all boils down to is, of course, *the sun*. (TED-ID40)

In this example, the post-copular constituent of the wh-cleft does not introduce new information but instead the exact words from the previous sentence are repeated, which are in this case *the sun.* This contradicts the assumption that the post-copular constituent always contains new information, or at least information that is less familiar to the hearer than the information in the wh-clause. I would argue that in this case the information *the sun* is certainly more present in the hearer´s consciousness than the information of the wh-clause. This example therefore does not confirm Prince's condition that the information in the post-copular constituent is always less present in the hearer's consciousness than the information in the wh-clause (Prince 1987: 892).

In two examples the post-copular constituent included both given and new information and were therefore coded as AMBIGUOUS (see 17).

17) But in the new project that I 'm working on, my new film, we examine the world of marketing, advertising. […] What I want to do is *make a film all about product placement, marketing and advertising*, where the entire film is funded by product placement, marketing and advertising. (TED-ID43)

This example shoes that the part of the post-copular constituent that is written in italics can be considered given information as the speaker already explicitly told the audience in the previous discourse that they are making a film on marketing and advertising. However, what is new to the audience is that this entire film will also be funded by product placement, marketing and advertising. For this reason, I

would argue that this post-copular constituent is both given and new and thus ambiguous.

GIVEN	INFERABLE	NEW	AMBIGUOUS
5	2	91	2
5%	2%	91%	2%

Table 3: Post-copular constituent information status corpus analysis results.

3.2.2 Type of Antecedents

The next category examined was the type of antecedent. In 69 examples (69%) the antecedent was linguistic in nature, with 21 explicit, 40 implicit and 8 contrastive antecedents in this category. There were also many examples in which the antecedents were either metalinguistic (15%) or situational (13%). The situational antecedents in particular can be attributed to the fact that the examples analyzed come from TED Talks, in which something is often presented visually or the information to which the wh-clause refers can be derived from something that is present in the speech situation. This means that not only the purely linguistic context is relevant but also the visual context. There were 3 examples in which the type of antecedent could not be clearly assigned to one of the categories (see 18).

18) But *I think* we can do both at the same time here. And we open up so many more possibilities. You can do so many more problems. *What I really think* we gain from this is students getting intuition and experience in far greater quantities than they 've ever got before. (TED-ID9)

In this excerpt, for example, it could on the one hand be argued that it is an explicit linguistic antecedent, since the speaker has already said that he is thinking something and then repeats this in the wh-clause. However, it could also be argued that it is a metalinguistic antecedent because from the metalinguistic notion that people think when they speak, it can be assumed that the question of what the speaker is thinking is present in the audience’s consciousness.

LINGUISTIC	METALINGUISTIC	SITUATIONAL	AMBIGUOUS
69	15	13	3
69%	15%	13%	3%

Table 4: Type of antecedent corpus analysis results.

3.2.3 Discourse Relations

In connection with the various types of antecedents, the following results were obtained for discourse relations (see table 5). In addition to the four classes provided in the Penn Discourse TreeBank 2.0, I added one more category of discourse relation, which is named *initial.* This category was added because in some of the TED-talk examples the wh-cleft occurred discourse initially or at least nearly discourse initially after a break with applause. This means that in these cases the wh-clause had no actual relation to the previous context but instead is used for example to introduce the topic of the talk. 3% of the wh-clauses from the analysis fell into this category. This is more than expected because from the notion that the information of wh-clauses is usually given or inferable from the previous context, it is very unusual that they can also occur as discourse openers. However, as Prince already noted, wh-clefts can occur at the beginning of a discourse under special circumstances.

19) I will never abandon you. I will be with you through the end. " Thank you very much. (Applause) *What I want to talk about today* is one idea. (TED-ID100)

In this example the wh-clause occurs right after a pause and serves as an introduction of the talk. Since the audience knows that in a TED Talk the speaker is talking about something, this wh-cleft can occur in the initial position, because it would be appropriate to assume that the question what the speaker is talking about today is in the audience's consciousness when the speaker starts talking.

As in none of the examples the discourse relation was temporal, it can be argued that the function of wh-clefts is usually not to express temporal sequences. Rather, in most of the cases, they serve to further elaborate on information that has already been mentioned directly or indirectly. Therefor 72% of the examples fall

into the category *expansion*. Another function is that they can introduce contrary ideas and thus generate a comparison with the previous context. This was the case for 18% of the TED Talk examples. An example for a wh-cleft that serves as a comparison is illustrated in the following.

20) So you 've all seen that -- New York Times, you 've seen it. *What you may not be familiar with* is that this research was spearheaded by NFL wives who said, " Isn 't it weird that my 46-year-old husband is forever losing his keys? (TED-ID10)

This shows that in the preceding context of the wh-cleft it was talked about something that everybody has seen and thus is familiar to everybody. The wh-clause *what you may not be familiar with* then provides a useful transition to introduce a topic that is unfamiliar to the audience and thus, contrasts with the previous context.

TEMPORAL	COMPARISON	CONTINGENCY	EXPANSION	INITIAL
0	18	7	72	3
0%	18%	7%	72%	3%

Table 5: Discourse relation corpus analysis results

3.2.4 Cleft Types

A very clear picture has emerged with regard to the results of the cleft types (see Table 6). By far the most common post-copular constituents were noun phrases (41%) and complementizer phrases (43%). This suggests that, at least in spoken language such as TED Talks, NP-clefts and CP-clefts are the most common cleft types and VP-, TP- and GP-clefts only playing a minor role.

NP	CP	VP	TP	GP
41	43	10	3	3
41%	43%	10%	3%	3%

Table 6: Cleft type corpus analysis results

3.2.5 Heaviness

The last category that was analyzed was the weight distribution of the wh-clause and the post-copular constituent in terms of number of words. In accordance with the rule that heavier constituents are placed at the end of the sentence and lighter constituents tend to come at the beginning, in 72% of cases the post-copular constituent was heavier, and in 19% of cases the post-copular constituent and wh-clause were approximately equal in terms of heaviness. However, there were also 9 examples in which the wh-clause was heavier than the post-copular constituent. Since this makes up almost 10% of the examples, it is worth taking a closer look at these sentences.

21) What excites me about using microbes is their efficiency. (TED-ID51)
1 2 3 4 5 6 1 2

In this example the wh-clause contains 6 words and the post-copular constituent only 2 words. This, as well as the other eight examples, proves that in wh-clefts, the post-copular constituent is not always heavier and thus contradict the hypothesis. Nevertheless, the corpus examples show that in cases where the wh-clause is heavier, there is usually only a difference of 2 to a maximum of 5 words between the wh-clause and the post-copular constituent. However, if the post-copular constituent is heavier, the two sentence elements often differ more significantly from each other (see 22).

22) What we did was we actually used the biomaterial as a bridge so that the
1 2 3 1 2 3 4 5 6 7 8 9 10 11
cells in the organ could walk on that bridge, if you will, and help to bridge
15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
the gap to regenerate that tissue. (TED-ID34)
31 32 33 34 35 36

This example clearly shows that the post-copular constituent contains 33 words more than the wh-clause and is therefore considerably heavier. This is not the only example in which this is the case. In 9% of the cases the post-copular constituent was 20 or more words heavier and in 10% of the cases 30 or more words heavier

than the wh-clause. This can most likely be explained by the fact that more words are needed to present or execute new information, while fewer words are required for information that is already known (Prince 1978: 886). Since this applies to most post-copular constituents, they are also ultimately heavier in most cases.

One correlation that emerged from the examples in this corpus analysis is between the information status of the post-copular constituent and its heaviness. In all examples where the post-copular constituent was coded as GIVEN, the post-copular constituent was also either equal or less heavy than the wh-clause. This underlines the assumption that less words are required to present given information and that the information of the post-copular constituent correlates with its heaviness.

PCC	WH=PCC	WH
72	19	9
72%	19%	9%

Table 7: Heaviness corpus analysis results

4 Conclusion

Overall, the results of the corpus analysis largely confirm the assumption made by Prince and other linguists, that wh-clauses contain information which is given or inferable for the hearer and that post-copular constituents introduce new information. However, the claim that the post-copular constituent must always be less familiar than the wh-clause cannot be fully agreed with, as some examples contradict this claim. In some cases, the information of the post-copular constituent was actually more accessible and present in the hearer's consciousness than the information of the wh-clause. In terms of antecedents, it can be concluded that in most of the cases the wh-clauses relied on linguistic information anchors, but also in a significant number of examples on metalinguistic or situational antecedents. The situational antecedents can be explained by the fact that TED Talks often use more than one medium, providing not only linguistic but also sometimes visual context. It can therefore be assumed that this type of antecedent is less likely to be found in written contexts, in which no visual context is provided. Furthermore, the results of the analysis of discourse relations illustrated that wh-clefts mainly serve

to expand on a point previously mentioned or indicated in the discourse. Another important function is that wh-clefts can be used to introduce an idea that is contrastive to previous information. What was interesting is that in 3% of the examples the wh-cleft occurred as a discourse opener after a break, which demonstrates that wh-clefts are suitable for introducing the topic of a talk. In each of these examples, the condition that the question expressed in the wh-clause would be appropriate in the hearer's consciousness, was fulfilled. For the distribution of cleft types, a very clear result emerged indicating that NP-clefts and CP-clefts are the most frequent types of wh-clefts in TED Talks. This suggests that the other cleft-types such as VP-, TP-, or GP-clefts only play a subordinate role in this genre. Moreover, the analysis of heaviness in terms of number of words of the two constituents largely confirmed the expectation that post-copular constituents tend to be heavier than wh-clauses or are at least equally heavy. However, almost 10% of the examples do not comply with this principle of placing the heavier constituent at the end of the sentence. Therefore, this analysis illustrated that the principle of end-weight is not a rule that applies at all times but rather depends on discourse needs or the information status. A correlation that was observed between givenness and heaviness. In all examples in which the post-copular constituent contained given information this constituent was also not heavier than the wh-clause suggesting that information status and heaviness are linked to a certain extent. Finally, it must be noted that the present corpus analysis is limited in that only *what*-clefts were examined and thus not all types of wh-clefts could be fully covered. In addition to that, only a maximum of 5 preceding sentences could be displayed in the corpus search, which means that no comprehensive picture of the context could be obtained.

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Appendix

1

18078

Consider first the reality that people today , in some cultures , are making choices about their offspring . They 're , in some cultures , choosing to have more males than females . It 's not necessarily good for the society , but it 's what the individual and the family are choosing . Think also , if it were possible ever for you to choose , not simply to choose the sex of your child , but for you in your body to make the genetic adjustments that would cure or prevent diseases . **What if you could** make the genetic changes to eliminate diabetes or Alzheimer 's or reduce the risk of cancer or eliminate stroke ? Wouldn 't you want to make those changes in your genes ? If we look ahead , these kind of changes are going to be increasingly possible . The Human Genome Project started in 1990 , and it took 13 years . It cost 2.7 billion dollars .

➔ No wh-cleft

2

18473

Can you imagine something good coming out of tobacco ? These are all reality today , and in the future , will be evermore possible . Imagine then just two other little changes . You can change the cells in your body , but what if you could change the cells in your offspring ? **What if you could** change the sperm and the ova , or change the newly fertilized egg , and give your offspring a better chance at a healthier life -- eliminate the diabetes , eliminate the hemophilia , reduce the risk of cancer ? Who doesn 't want healthier children ? And then , that same analytic technology , that same engine of science that can produce the changes to prevent disease , will also enable us to adopt super-attributes , hyper-capacities -- that better memory . Why not have the quick wit of a Ken Jennings , especially if you can augment it with the next generation of the Watson machine ? Why not have the quick twitch muscle that will enable you to run faster and longer ?

➔ No wh-cleft

ID1:

21133

You might say , " Well , maybe things just smoothed themselves out . " But the conditions near the Big Bang are very , very different than the conditions of the air in this room . In particular , things were a lot denser . The gravitational pull of things was a lot stronger near the Big Bang . What you have to think about is we have a universe with a hundred billion galaxies , a hundred billion stars each . At early times , those hundred billion galaxies were squeezed into a region about this big -- literally , at early times . And you have to imagine doing that squeezing without any imperfections , without any little spots where there were a few more atoms than somewhere else . Because if there had been , they would have collapsed under the gravitational pull into a huge black hole . Keeping the universe very , very smooth at early times is not easy , it 's a delicate arrangement .

ID2:

24997

You can imagine that then maybe you an use a pure protein to guide light , and so we 've made optical fibers . But silk is versatile and it goes beyond optics . And you can think of different formats . So for instance , if you 're afraid of going to the doctor and getting stuck with a needle , we do microneedle arrays . What you see there on the screen is a human hair superimposed on the needle that 's made of silk -- just to give you a sense of size . You can do bigger things . You can do gears and nuts and bolts -- that you can buy at Whole Foods . And the gears work in water as well . So you think of alternative mechanical parts .

ID3:

27716

And without skipping a beat at all , he said , " Well , they 'd be about the same . The only difference is a Susskind sandwich would have a lot more ham , " ham , as in bad actor . (Laughter) Well , I happened to have been very quick that day , and I said , " Yeah , but a lot less baloney . " (Laughter) The truth of the matter is that a Feynman sandwich had a load of ham , but absolutely no baloney . What Feynman hated worse than anything else was intellectual pretense -- phoniness , false sophistication , jargon . I remember sometime during the 80s , the mid-80s , Dick

and I and Sidney Coleman , we met a couple of times up in San Francisco at some very rich guy 's house -- up in San Francisco for dinner . And the last time the rich guy invited us , he also invited a couple of philosophers . These guys were philosophers of mind . Their specialty was the philosophy of consciousness .

ID4:

35105

This film tried to find a balance between telling a political story , but also a feminine story . Being a visual artist , indeed , I am foremost interested to make art -- to make art that transcends politics , religion , the question of feminism , and become an important , timeless , universal work of art . The challenge I have is how to do that -- how to tell a political story by an allegorical story -- how to move you with your emotions , but also make your mind work . These are some of the images and the characters of the film . What is unbelievably ironic is the period that we tried to depict in the film , the cry for democracy and social justice , repeats itself now again in Tehran . The green movement significantly inspired the world . It brought a lot of attention to all those Iranians who stand for basic human rights and struggle for democracy . What was most significant for me was , once again , the presence of the women . They 're absolutely inspirational for me .

ID5:

These are some of the images and the characters of the film . What is unbelievably ironic is the period that we tried to depict in the film , the cry for democracy and social justice , repeats itself now again in Tehran . The green movement significantly inspired the world . It brought a lot of attention to all those Iranians who stand for basic human rights and struggle for democracy . What was most significant for me was , once again , the presence of the women . They 're absolutely inspirational for me . If in the Islamic Revolution , the images of the woman portrayed were submissive and didn 't have a voice , now we saw a new idea of feminism in the streets of Tehran -- women who were educated , forward thinking , non-traditional , sexually open , fearless and seriously feminist . These women and those young men united Iranians across the world , inside and outside . I then discovered why I take so much inspiration from Iranian women .

ID6:

38423

So in 2007 , our team won half a million dollars by placing third place in this competition . So about that time , the National Federation of the Blind , or NFB , challenged the research committee about who can develop a car that lets a blind person drive safely and independently . We decided to give it a try , because we thought , hey , how hard could it be . We have already an autonomous vehicle . What NFB wanted was not a vehicle that can drive a blind person around , but a vehicle where a blind person can make active decisions and drive . So we had to throw everything out the window and start from scratch . So to test this crazy idea , we developed a small dune buggy prototype vehicle to test the feasibility . It was an absolutely amazing experience . But the problem with this car was it was designed to only be driven in a very controlled environment , in a flat closed-off parking lot -- even the lanes defined by red traffic cones .

3

44943

Many of these are a father who has to leave his family behind to go get a job -- sometimes a single mother who has to get enough money to even pay for food or clothing . Sadly , sometimes there are even children who are trying to get reunited with their parents that they haven 't seen , in some cases , for a decade . So what kind of an idea is it to think about building a brand new city in Honduras ? Or to build a dozen of these , or a hundred of these around the world ? What kind of an idea is it to think about insisting that every family have a choice of several cities that are competing to attract new residents ? This is an idea worth spreading . And my friends from Honduras asked me to say thank you , TED . (Applause) From all outward appearances , John had everything going for him .

➔ No wh-cleft

ID7:

51794

Like for the hamburger , we just cut the damn barbecue . And so those coals that kept falling off the edge , we kept having to put them back up . But again , it only has to work for a thousandth of a second . The wok shot caught fire three times . What happens when you have your wok cut in half is the oil goes down into the fire and whoosh ! One of our cooks lost his eyebrows that way . But hey , they grow back . In addition to cutaways , we also explain physics . This is Fourier 's law of heat conduction .

ID8:

60394

Look at the health systems in so many countries . No . Could the private sector and NGOs get together and marshal a response ? Absolutely . What they lack is that safe space to come together , agree and move to action . That 's what we 're trying to provide . I know as well that this often seems like an overwhelming level of responsibility for people to assume . " You want me to deliver human rights throughout my global supply chain . There are thousands of suppliers in there . "

ID9:

63802

It 's usually some kind of choice between the vocational and the intellectual . But I think we can do both at the same time here . And we open up so many more possibilities . You can do so many more problems . What I really think we gain from this is students getting intuition and experience in far greater quantities than they 've ever got before . And experience of harder problems -- being able to play with the math , interact with it , feel it . We want people who can feel the math instinctively . That 's what computers allow us to do . Another thing it allows us to do is reorder the curriculum .

ID10:

71320

And here 's the really alarming part : we don 't fully understand the long-term impact of multiple injuries . You guys may be familiar with this research that 's coming out of the NFL . In a nutshell , this research suggests that among retired NFL players with three or more career concussions , the incidents of early-onset dementing disease is much greater than it is for the general population . So you 've all seen that -- New York Times , you 've seen it . What you may not be familiar with is that this research was spearheaded by NFL wives who said , " Isn 't it weird that my 46-year-old husband is forever losing his keys ? Isn 't it weird that my 47-year-old husband is forever losing the car ? So it 's going to be really important that he be able to drive me around some day . So how do we guarantee the safety of our kids ? How can we 100 percent guarantee the safety of our kids ?

4

76057

So let 's go through a couple of those . Now , managers and bosses will often have you think that the real distractions at work are things like Facebook and Twitter and YouTube and other websites , and in fact , they 'll go so far as to actually ban these sites at work . Some of you may work at places where you can 't get to these certain sites . I mean , is this China ? What the hell is going on here ? You can 't go to a website at work , and that 's the problem , that 's why people aren 't getting work done , because they 're going to Facebook and they 're going to Twitter ? That 's kind of ridiculous . It 's a total decoy . And today 's Facebook and Twitter and YouTube , these things are just modern-day smoke breaks .

➔ No wh-cleft

ID11:

77945

It seems to me like corporations are always trying to get kids , like me , to get their parents to buy stuff that really isn 't good for us or the planet . Little kids , especially , are attracted by colorful packaging and plastic toys . I must admit , I used to be one of them . I also used to think that all of our food came from these happy , little

farms where pigs rolled in mud and cows grazed on grass all day . What I discovered was this is not true . I began to look into this stuff on the Internet , in books and in documentary films , in my travels with my family . I discovered the dark side of the industrialized food system . First , there 's genetically engineered seeds and organisms . Don 't get me wrong , I like fish and tomatoes , but this is just creepy .

5

80702

Terrorism is basically taking an innocent stranger and treating them as an enemy whom you kill in order to create fear . What 's the opposite of terrorism ? It 's taking an innocent stranger and treating them as a friend whom you welcome into your home in order to sow and create understanding , or respect , or love . So what if then you took the story of Abraham , which is a third side story , what if that could be -- because Abraham stands for hospitality -- what if that could be an antidote to terrorism ? What if that could be a vaccine against religious intolerance ? How would you bring that story to life ? Now it 's not enough just to tell a story -- that 's powerful -- but people need to experience the story . They need to be able to live the story . How would you do that ?

➔ No wh-cleft

ID12:

88426

So you 've got to watch your space for that . Five weeks old , it 's my baby , and it 's hurting real bad . The People 's Supermarket . So basically , the restaurants only really hit people who believed in what I was doing anyway . What I needed to do was get food out to a broader spectrum of people . This is a social enterprise , not-for-profit cooperative supermarket . It really is about the social disconnect between food , communities in urban settings and their relationship to rural growers -- connecting communities in London to rural growers . Really important . So I 'm committing to potatoes ; I 'm committing to milk ; I 'm committing to leeks and broccoli -- all very important stuff .

ID13:

92165

How many still don 't see it ? How many engines on the wing of that Boeing ? (Laughter) Right in the middle of the picture ! Thanks very much for your attention . What I wanted to show you is that scientists , using their from-the-outside , third-person methods , can tell you things about your own consciousness that you would never dream of , and that , in fact , you 're not the authority on your own consciousness that you think you are . And we 're really making a lot of progress on coming up with a theory of mind . Jeff Hawkins , this morning , was describing his attempt to get theory , and a good , big theory , into the neuroscience . And he 's right . This is a problem .

ID14:

98847

We used to have a manufacturing economy , which was about building goods and products , and now we have a service economy and an information and creative economy . It used to be that you were a guy who went to high school who didn 't have a college degree , but you had a specific set of skills , and with the help of a union , you could make yourself a pretty good middle-class life . But that really isn 't true anymore . This new economy is pretty indifferent to size and strength , which is what 's helped men along all these years . What the economy requires now is a whole different set of skills . If you look at management theory these days , it used to be that our ideal leader sounded something like General Patton , right ? You would be issuing orders from above . You would be very hierarchical . You would tell everyone below you what to do .

ID15:

112657

What 's the theme ? What 's the pattern ? My husband left town with the kids because I always go into this Jackson Pollock crazy thing , where I 'm just like writing and in my researcher mode . And so here 's what I found . What they had in common was a sense of courage . And I want to separate courage and bravery for you for a minute . Courage , the original definition of courage , when it first came into the English language -- it 's from the Latin word cor , meaning heart -- and the

original definition was to tell the story of who you are with your whole heart . And so these folks had , very simply , the courage to be imperfect . They had the compassion to be kind to themselves first and then to others , because , as it turns out , we can 't practice compassion with other people if we can 't treat ourselves kindly .

ID16:
123265
I will do my best this morning to tell you what I think is the truth . But first , my disclosures . I am not a breast cancer survivor . I 'm not a radiologist . (Laughter) What I am is a doctor of internal medicine who became passionately interested in this topic about 10 years ago when a patient asked me a question . She came to see me after discovering a breast lump . Her sister had been diagnosed with breast cancer in her 40s . She and I were both very pregnant at that time , and my heart just ached for her , imagining how afraid she must be . Fortunately , her lump proved to be benign .

6
141958
Given the stakes , the climate crisis clearly calls for us to act based on the precautionary principle -- the theory that holds that when human health and the environment are significantly at risk and when the potential damage is irreversible , we cannot afford to wait for perfect scientific certainty . Better to err on the side of caution . More overt , the burden of proving that a practice is safe should not be placed on the public that would be harmed , but rather on the industry that stands to profit . But climate policy in the wealthy world -- to the extent that such a thing exists -- is not based on precaution , but rather on cost-benefit analysis -- finding the course of action that economists believe will have the least impact on our GDP . So rather than asking , as precaution would demand , what can we do as quickly as possible to avoid potential catastrophe , we ask bizarre questions like this : " What is the latest possible moment we can wait before we begin seriously lowering emissions ? It 's coming from the economists imposing their mechanistic thinking on the science . The fact is that we simply don 't know when the warming that we create will be utterly overwhelmed by feedback loops . So once again , why do we

take these crazy risks with the precious ? A range of explanations may be popping into your mind by now , like " greed . "

➔ No we-cleft

ID17:

149356

And I 'll tell you now , that East Asianers , people from East Asia , are far more knowledgeable about the West than the West is about East Asia . Now this point is very germane , I 'm afraid , to the present . Because what 's happening ? Back to that chart at the beginning , the Goldman Sachs chart . What is happening is that , very rapidly in historical terms , the world is being driven and shaped , not by the old developed countries , but by the developing world . We 've seen this in terms of the G20 usurping very rapidly the position of the G7 , or the G8 . And there are two consequences of this . First , the West is rapidly losing its influence in the world . There was a dramatic illustration of this actually a year ago -- Copenhagen , climate change conference .

ID18:

155198

They go right to doctors . And God forbid -- I think many doctors , if you really asked them , they don 't really understand all this stuff either . This is the worst presented information . You ask Tufte , and he would say , " Yes , this is the absolute worst presentation of information possible . " What we did at Wired was we went , and I got our graphic design department to re-imagine these lab reports . So that 's what I want to walk you through . So this is the general blood work before , and this is the after , this is what we came up with . The after takes what was four pages -- that previous slide was actually the first of four pages of data that 's just the general blood work . It goes on and on and on , all these values , all these numbers you don 't know .

7

162512

But the advantage of that little bit of social discomfort is the fact that , when it comes to running under hot heat for long distances , we 're superb , we 're the best on the planet . You take a horse on a hot day , and after about five or six miles , that horse has a choice . It 's either going to breathe or it 's going to cool off , but it ain 't doing both -- we can . So what if we evolved as hunting pack animals ? What if the only natural advantage we had in the world was the fact that we could get together as a group , go out there on that African Savannah , pick out an antelope and go out as a pack and run that thing to death ? Well if that 's true , a couple other things had to be true as well . The key to being part of a hunting pack is the word " pack . " If you go out by yourself , and you try to chase an antelope , I guarantee you there 's going to be two cadavers out there in the Savannah . You need a pack to pull together .

➔ No wh-cleft

ID19:

164101

I 've been wrong a lot lately . Like a lot . So I couldn 't figure out what to read today . I mean , I 've been saying I 've been prepping . What that means is prepping my outfit , (Laughter) prepping options , trying to figure out what I 'm coming behind and going in front of . Poetry does that . It preps you . It aims you . So I am going to read a poem that was chosen just now .

ID20:

164596

Do not fear what has blown up . If you must , fear the unexploded . Thank you . (Applause) What I thought I would do is I would start with a simple request . I 'd like all of you to pause for a moment , you wretched weaklings , and take stock of your miserable existence . It was the advice that I decided to follow myself when I turned 40 . Up until that moment , I had been that classic corporate warrior -- I was eating too much , I was drinking too much , I was working too hard and I was neglecting the family . And I decided that I would try and turn my life around .

ID21:

168760

I think that 's very , very important . So this is a new project . We 've invited a lot of kids into this space , and they think it 's pretty cool . But I can tell you , the thing that they love the most is the robot . What they care about is the robot . Robots touch something deeply human within us . And so whether they 're helping us to become creative and innovative , or whether they 're helping us to feel more deeply connected despite distance , or whether they are our trusted sidekick who 's helping us attain our personal goals in becoming our highest and best selves , for me , robots are all about people . Thank you . (Applause)

ID22:

174117

What you 're drawn to are her eyes and the skin you love to touch . But today I 'm going to talk to you about something you can 't see -- what 's going on up in that little brain of hers . The modern tools of neuroscience are demonstrating to us that what 's going on up there is nothing short of rocket science . And what we 're learning is going to shed some light on what the romantic writers and poets described as the " celestial openness " of the child 's mind . What we see here is a mother in India , and she 's speaking Koro , which is a newly discovered language . And she 's talking to her baby . What this mother -- and the 800 people who speak Koro in the world -- understands [is] that , to preserve this language , they need to speak it to the babies . And therein lies a critical puzzle . Why is it that you can 't preserve a language by speaking to you and I , to the adults ?

ID23:

174222

What this mother -- and the 800 people who speak Koro in the world -- understands [is] that , to preserve this language , they need to speak it to the babies . And therein lies a critical puzzle . Why is it that you can 't preserve a language by speaking to you and I , to the adults ? Well , it 's got to do with your brain . What we see here is that language has a critical period for learning . The way to read this slide is to look at your age on the horizontal axis . (Laughter) And you 'll see on the vertical your skill at acquiring a second language . Babies and children are geniuses until

they turn seven , and then there 's a systematic decline . After puberty , we fall off the map .

ID24:
174594
We 're culture-bound listeners . We can discriminate the sounds of our own language , but not those of foreign languages . So the question arises : when do those citizens of the world turn into the language-bound listeners that we are ? And the answer : before their first birthdays . What you see here is performance on that head-turn task for babies tested in Tokyo and the United States , here in Seattle , as they listened to " ra " and " la " -- sounds important to English , but not to Japanese . So at six to eight months the babies are totally equivalent . Two months later something incredible occurs . The babies in the United States are getting a lot better , babies in Japan are getting a lot worse , but both of those groups of babies are preparing for exactly the language that they are going to learn . So the question is : what 's happening during this critical two-month period ?

ID25:
175138
We knew that , when monolinguals were tested in Taipei and Seattle on the Mandarin sounds , they showed the same pattern . Six to eight months , they 're totally equivalent . Two months later , something incredible happens . But the Taiwanese babies are getting better , not the American babies . What we did was expose American babies during this period to Mandarin . It was like having Mandarin relatives come and visit for a month and move into your house and talk to the babies for 12 sessions . Here 's what it looked like in the laboratory . (Video) Mandarin Speaker : [Mandarin] PK : So what have we done to their little brains ? (Laughter) We had to run a control group to make sure that just coming into the laboratory didn 't improve your Mandarin skills .

ID26:
175295
So a group of babies came in and listened to English . And we can see from the graph that exposure to English didn 't improve their Mandarin . But look at what

happened to the babies exposed to Mandarin for 12 sessions . They were as good as the babies in Taiwan who 'd been listening for 10-and-a-half months . What it demonstrated is that babies take statistics on a new language . Whatever you put in front of them , they 'll take statistics on . But we wondered what role the human being played in this learning exercise . So we ran another group of babies in which the kids got the same dosage , the same 12 sessions , but over a television set and another group of babies who had just audio exposure and looked at a teddy bear on the screen . What did we do to their brains ?

ID27:

175390

Whatever you put in front of them , they 'll take statistics on . But we wondered what role the human being played in this learning exercise . So we ran another group of babies in which the kids got the same dosage , the same 12 sessions , but over a television set and another group of babies who had just audio exposure and looked at a teddy bear on the screen . What did we do to their brains ? What you see here is the audio result -- no learning whatsoever -- and the video result -- no learning whatsoever . It takes a human being for babies to take their statistics . The social brain is controlling when the babies are taking their statistics . We want to get inside the brain and see this thing happening as babies are in front of televisions , as opposed to in front of human beings . Thankfully , we have a new machine , magnetoencephalography , that allows us to do this .

8

176114

These conversations really reflect what 's happening at the national and international level . One of my favorite quotes from literature was written by Tillie Olsen , the great American writer from the South . " Better immersion than to live untouched . " But the real question is , what is the cost of not daring ? What is the cost of not trying ? I 've been so privileged in my life to know extraordinary leaders who have chosen to live lives of immersion . One woman I knew who was a fellow at a program that I ran at the Rockefeller Foundation was named Ingrid Washinawatok . She was a leader of the Menominee tribe , a Native American

peoples . And when we would gather as fellows , she would push us to think about how the elders in Native American culture make decisions .

➔ No wh-cleft

ID28:

177275

And I have learned so much from Chris and Kevin and Alex and Herbert and all of these young men . Alex , in some ways , said it best . He said , " We used to feel like nobodies , but now we feel like somebodies . " And I think we have it all wrong when we think that income is the link . What we really yearn for as human beings is to be visible to each other . And the reason these young guys told me that they 're doing these TEDx 's is because they were sick and tired of the only workshops coming to the slums being those workshops focused on HIV , or at best , microfinance . And they wanted to celebrate what 's beautiful about Kibera and Mathare -- the photojournalists and the creatives , the graffiti artists , the teachers and the entrepreneurs . And they 're doing it . And my hat 's off to you in Kibera .

ID29:

181517

It 's now traveling . At the moment it 's in Indianapolis , but it was at the Smithsonian . I remember when you were the first woman as Secretary of State , and there was a lot of conversation always about what you were wearing , how you looked -- the thing that happens to a lot of women , especially if they 're the first in a position . But people did pay attention to what clothes I had . What was interesting was that , before I went up to New York as U.N. ambassador , I talked to Jeane Kirkpatrick , who 'd been ambassador before me , and she said , " You 've got to get rid of your professor clothes . Go out and look like a diplomat . " So that did give me a lot of opportunities to go shopping . And one of the things -- if you remember Condoleezza Rice was at some event and she wore boots , and she got criticized over that . And no guy ever gets criticized .

9

181897

And all of a sudden I thought , " Well , wait a minute . And so that happened more at various times , but I really think that there was a great advantage in many ways to being a woman . I think we are a lot better at personal relationships , and then have the capability obviously of telling it like it is when it 's necessary . (Laughter) (Applause) PM : Because in her lifetime -- MA : That would be so . PM : What a change that is . As you travel now all over the world , which you do frequently , how do you assess this global narrative around the story of women and girls ? And therefore it means that we have to remember that , while many of us have had huge opportunities -- and Pat , you have been a real leader in your field -- is that there are a lot of women that are not capable of worrying and taking care of themselves and understanding that women have to help other women . And so what I have felt -- and I have looked at this from a national security issue -- when I was Secretary of State , I decided that women 's issues had to be central to American foreign policy , not just because I 'm a feminist , but because I believe that societies are better off when women are politically and economically empowered , that values are passed down , the health situation is better , education is better , there is greater economic prosperity . So I think that it behooves us -- those of us that live in various countries where we do have economic and political voice -- that we need to help other women .

➔ No wh-cleft

10

184588

First , we 've got to be ready and willing to take experts on and dispense with this notion of them as modern-day apostles . This doesn 't mean having to get a Ph.D. in every single subject , you 'll be relieved to hear . But it does mean persisting in the face of their inevitable annoyance when , for example , we want them to explain things to us in language that we can actually understand . You see , being ready to take experts on is about also being willing to dig behind their graphs , their equations , their forecasts , their prophecies , and being armed with the questions to do that -- questions like : What are the assumptions that underpin this ? What is the evidence upon which this is based ? What has your investigation focused on ? And

what has it ignored ? It recently came out that experts trialing drugs before they come to market typically trial drugs first , primarily on male animals and then , primarily on men . It seems that they 've somehow overlooked the fact that over half the world 's population are women .

➔ No wh-cleft

ID30:

186613

Well that 's not surprising , but we also asked them to judge them on honesty , intelligence , friendliness , violence . After the surgery , they were perceived as being more intelligent , more friendly , more honest , less violent -- and yet we hadn 't operated on their intellect or their character . When people get older , they don 't necessarily choose to follow this kind of surgery . Their presence in the consultation suite is a result of the slings and arrows of outrageous fortune . What happens to them is that they may have suffered cancer or trauma . So this is a photograph of Henry , two weeks after he had a malignant cancer removed from the left side of his face -- his cheekbone , his upper jaw , his eye-socket . He looks pretty good at this stage . But over the course of the next 15 years he had 14 more operations , as the disease ravaged his face and destroyed my reconstruction regularly . He worked as an advocate .

ID31:

189623

(Applause) (Music) (Applause) Angella Ahn : Thank you . (Applause) Thank you so much . We are so honored to be here at TEDWomen , sharing our music with you . What an exciting and inspiring event . What you just heard is " Skylife " by David Balakrishnan . We want to play you one more selection . It 's by Astor Piazzolla , an Argentine composer . And we talk about different ideas -- he had this idea that he thought music should be from the heart . It was more atonal and twelve-tone .

ID32:

191628

I took more photos , and we started the project . The drug lords were kind of worried about us filming in the place , so I told them , " You know what ? I 'm not interested in filming the violence and the weapons . You see that enough in the media . What I want to show is the incredible life . And actually I 've been seeing it around me the last few days . " So that 's a really symbolic pasting , because that 's the first one we did that you couldn 't see from the city . And that 's where the three kids got arrested , and that 's the grandmother of one of them . And on that stairs , that 's where the traffickers always stand and there 's a lot of exchange of fire .

ID33:

191739

And that 's where the three kids got arrested , and that 's the grandmother of one of them . And on that stairs , that 's where the traffickers always stand and there 's a lot of exchange of fire . Everyone there understood the project . And then we pasted everywhere -- the whole hill . (Applause) What was interesting is that the media couldn 't get in . I mean , you should see that . They would have to film us from a really long distance by helicopter and then have a really long lens , and we would see ourselves , on TV , pasting . And they would put a number : " Please call this number if you know what 's going on in Providencia . " We just did a project and then left so the media wouldn 't know .

ID34:

197671

You saw the spray going in . That was like the fibers of the cotton candy creating this structure , this tubularized structure , which is a biomaterial that we can then use to help your body regenerate using your very own cells to do so . And that 's exactly what we did here . This is actually a patient who [was] presented with a deceased organ , and we then created one of these smart biomaterials , and then we then used that smart biomaterial to replace and repair that patient 's structure . What we did was we actually used the biomaterial as a bridge so that the cells in the organ could walk on that bridge , if you will , and help to bridge the gap to regenerate that tissue . And you see that patient now six months after with an X-ray showing you

the regenerated tissue , which is fully regenerated when you analyze it under the microscope . We can also use cells alone . These are actually cells that we obtained . These are stem cells that we create from specific sources , and we can drive them to become heart cells , and they start beating in culture .

ID35:
201443
Instead , what she really wanted to do was make films . So she made a film about the welfare system and had a huge impact . I wrote about Maricela Guzman , the daughter of Mexican immigrants , who joined the military so she could afford college . She was actually sexually assaulted in boot camp and went on to co-organize a group called the Service Women 's Action Network . What I learned from these people and others was that I couldn 't judge them based on their failure to meet their very lofty goals . Many of them are working in deeply intractable systems -- the military , congress , the education system , etc . But what they managed to do within those systems was be a humanizing force . And at the end of the day , what could possibly be more important than that ? Cornel West says , " Of course it 's a failure .

ID36:
204894
And so what I 'm showing you here , this is actually data from a pilot in the Los Altos school district , where they took two fifth grade classes and two seventh grade classes and completely gutted their old math curriculum . These kids aren 't using textbooks , they 're not getting one-size-fits-all lectures . They 're doing Khan Academy , they 're doing that software , for roughly half of their math class . And I want to make it clear , we don 't view this as the complete math education . What it does is -- and this is what 's happening in Los Altos -- it frees up time . This is the blocking and tackling , making sure you know how to move through a system of equations , and it frees up time for the simulations , for the games , for the mechanics , for the robot building , for the estimating how high that hill is based on its shadow . And so the paradigm is the teacher walks in every day , every kid works at their own pace -- and this is actually a live dashboard from Los Altos school district --

and they look at this dashboard . Every row is a student . Every column is one of those concepts .

ID37:

207622

So one of the problems with the filter bubble was discovered by some researchers at Netflix . And they were looking at the Netflix queues , and they noticed something kind of funny that a lot of us probably have noticed , which is there are some movies that just sort of zip right up and out to our houses . They enter the queue , they just zip right out . So " Iron Man " zips right out , and " Waiting for Superman " can wait for a really long time . What they discovered was that in our Netflix queues there 's this epic struggle going on between our future aspirational selves and our more impulsive present selves . You know we all want to be someone who has watched " Rashomon , " but right now we want to watch " Ace Ventura " for the fourth time . (Laughter) So the best editing gives us a bit of both . It gives us a little bit of Justin Bieber and a little bit of Afghanistan . It gives us some information vegetables ; it gives us some information dessert .

ID38:

207779

It gives us a little bit of Justin Bieber and a little bit of Afghanistan . It gives us some information vegetables ; it gives us some information dessert . And the challenge with these kinds of algorithmic filters , these personalized filters , is that , because they 're mainly looking at what you click on first , it can throw off that balance . And instead of a balanced information diet , you can end up surrounded by information junk food . What this suggests is actually that we may have the story about the Internet wrong . In a broadcast society -- this is how the founding mythology goes -- in a broadcast society , there were these gatekeepers , the editors , and they controlled the flows of information . And along came the Internet and it swept them out of the way , and it allowed all of us to connect together , and it was awesome . But that 's not actually what 's happening right now . What we 're seeing is more of a passing of the torch from human gatekeepers to algorithmic ones .

ID39:

216630

It was Einstein 's great general theory of relativity . Now even light will be bent by those paths . And you can be bent so much that you 're caught in orbit around the Sun , as the Earth is , or the Moon around the Earth . These are the natural curves in space . What Einstein did not realize was that , if you took our Sun and you crushed it down to six kilometers -- so you took a million times the mass of the Earth and you crushed it to six kilometers across , you would make a black hole , an object so dense that if light veered too close , it would never escape -- a dark shadow against the universe . It wasn 't Einstein who realized this , it was Karl Schwarzschild who was a German Jew in World War I -- joined the German army already an accomplished scientist , working on the Russian front . I like to imagine Schwarzschild in the war in the trenches calculating ballistic trajectories for cannon fire , and then , in between , calculating Einstein 's equations -- as you do in the trenches . And he was reading Einstein 's recently published general theory of relativity , and he was thrilled by this theory . And he quickly surmised an exact mathematical solution that described something very extraordinary : curves so strong that space would rain down into them , space itself would curve like a waterfall flowing down the throat of a hole .

ID40:

220314

If we would apply these codes and standards to the Pantheon in Rome , it would never have looked like this , because this beautiful light feature that goes around there all by itself can only appear because there is also darkness in that same building . And the same is more or less what Santiago Calatrava said when he said , " Light : I make it in my buildings for comfort . " And he didn 't mean the comfort of a five-course dinner as opposed to a one-course meal , but he really meant the comfort of the quality of the building for the people . He meant that you can see the sky and that you can experience the sun . What it all boils down to is , of course , the sun . And this image of the sun may suggest that the sun is something evil and aggressive , but we should not forget that all energy on this planet actually comes from the sun , and light is only a manifestation of that energy . The sun is for dynamics , for color changes . The sun is for beauty in our environment , like in this

building -- the High Museum in Atlanta , which has been created by Renzo Piano from Italy , together with Arup Lighting , a brilliant team of lighting designers , who created a very subtle modulation of light across the space , responding to what the sun does outside , just because of all these beautiful openings in the roof . So in an indirect way , you can see the sun .

ID41:

227970

First , population growth . Second , economic growth . Population growth will mainly occur among the poorest people here because they have high child mortality and they have many children per woman . And [with] that you will get two extra , but that won 't change the energy use very much . What will happen is economic growth . The best of here in the emerging economies -- I call them the New East -- they will jump the air line . " Wopp ! " they will say . And they will start to use as much as the Old West are doing already . And these people , they want the washing machine .

ID42:

233293

And that worked pretty well . So we were able to associate the small digital warhead with the rotor control . The rotor is that moving part within the centrifuge , that black object that you see . And if you manipulate the speed of this rotor , you are actually able to crack the rotor and eventually even have the centrifuge explode . What we also saw is that the goal of the attack was really to do it slowly and creepy -- obviously in an effort to drive maintenance engineers crazy , that they would not be able to figure this out quickly . The big digital warhead -- we had a shot at this by looking very closely at data and data structures . So for example , the number 164 really stands out in that code ; you can 't overlook it . So that made sense , that was a match . And it even got better .

11

240904

Well it turns out there is a way to measure happiness in the brain . And you do that by measuring the relative activation of the left prefrontal cortex in the fMRI , versus

the right prefrontal cortex . And Matthieu 's happiness measure is off the charts . He 's by far the happiest man ever measured by science . Which leads us to a question : What was he thinking when he was being measured ? Perhaps something very naughty . (Laughter) Actually , he was meditating on compassion . Matthieu 's own experience is that compassion is the happiest state ever . Reading about Matthieu was one of the pivotal moments of my life .

➔ No wh-cleft

12

241101

Because if compassion was a chore , nobody 's going to do it , except maybe the Dalai Lama or something . But if compassion was fun , everybody 's going to do it . Therefore , to create the conditions for global compassion , all we have to do is to reframe compassion as something that is fun . But fun is not enough . What if compassion is also profitable ? What if compassion is also good for business ? Then , every boss , every manager in the world , will want to have compassion -- like this . That would create the conditions for world peace . So , I started paying attention to what compassion looks like in a business setting .

➔ No wh-cleft

13

241108

But if compassion was fun , everybody 's going to do it . Therefore , to create the conditions for global compassion , all we have to do is to reframe compassion as something that is fun . But fun is not enough . What if compassion is also profitable ? What if compassion is also good for business ? Then , every boss , every manager in the world , will want to have compassion -- like this . That would create the conditions for world peace . So , I started paying attention to what compassion looks like in a business setting . Fortunately , I didn 't have to look very far .

➔ No wh-cleft

ID43:

243333

Now if you 'd have asked me that a year ago , I wouldn 't have been able to tell you that with any certainty . But in the new project that I 'm working on , my new film , we examine the world of marketing , advertising . And as I said earlier , I put myself in some pretty horrible situations over the years , but nothing could prepare me , nothing could ready me , for anything as difficult or as dangerous as going into the rooms with these guys . (Laughter) You see , I had this idea for a movie . (Video) Morgan Spurlock : What I want to do is make a film all about product placement , marketing and advertising , where the entire film is funded by product placement , marketing and advertising . So the movie will be called " The Greatest Movie Ever Sold . " And so the film explores this whole idea -- (Michael Kassan : It 's redundant .) In perpetuity , forever ? I 'm a redundant person .

14

244139

But why ? Well , transparency is scary -- (Roar) like that odd , still-screaming bear . (Laughter) It 's unpredictable -- (Music) (Laughter) like this odd country road . And it 's also very risky . (Laughter) What else is risky ? Eating an entire bowl of Cool Whip . (Laughter) That 's very risky . Now when I started talking to companies and telling them that we wanted to tell this story , and they said , " No , we want you to tell a story . We want you to tell a story , but we just want to tell our story . "

➔ No wh-cleft

ID44:

246360

It 's not even streaming . It 's not even been out into other foreign countries yet . So ultimately , this film has already started to gain a lot of momentum . And not bad for a project that almost every ad agency we talked to advised their clients not to take part . What I always believe is that if you take chances , if you take risks , that in those risks will come opportunity . I believe that when you push people away from that , you 're pushing them more towards failure . I believe that when you train your employees to be risk averse , then you 're preparing your whole company to

be reward challenged . I feel like that what has to happen moving forward is we need to encourage people to take risks . We need to encourage people to not be afraid of opportunities that may scare them .

ID45:

257055

Heart disease , stroke , diabetes , spinal cord injury , muscular dystrophy , retinal eye diseases -- are any of these conditions relevant , personally , to you ? I was about to start a robotic operation , but stepping out of the elevator into the bright and glaring lights of the operating room , I realized that my left visual field was fast collapsing into darkness . Earlier that week , I had taken a rather hard knock during late spring skiing -- yes , I fell . And I started to see floaters and stars , which I casually dismissed as too much high-altitude sun exposure . What happened to me might have been catastrophic , if not for the fact that I was in reach of good surgical access . And I had my vision restored , but not before a prolonged period of convalescence -- three months -- in a head down position . This experience taught me to empathize more with my patients , and especially those with retinal diseases . 37 million people worldwide are blind , and 127 million more suffer from impaired vision . Stem cell-derived retinal transplants , now in a research phase , may one day restore vision , or part vision , to millions of patients with retinal diseases worldwide .

ID46:

264822

So we made a film and it turns out to be more than a story about me , more than a story about one teacher . It 's a story that 's a testament to teaching and teachers . And it 's a beautiful thing . And the strange thing is , when I watch the film -- I have the eerie sensation of seeing it -- I saw myself literally disappear . What I saw was my teachers coming through me . I saw my geometry teacher in high school , Mr. Rucell 's wry smile under his handlebar mustache . That 's the smile I use -- that 's his smile . I saw Jan Polo 's flashing eyes . And they weren 't flashing in anger , they were flashing in love , intense love for her students .

15

270256

Consider , first , the reality that people today , in some cultures , are making choices about their offspring . They 're , in some cultures , choosing to have more males than females . It 's not necessarily good for the society , but it 's what the individual and the family are choosing . Think also , if it were possible ever for you to choose , not simply to choose the sex of your child , but for you in your body to make the genetic adjustments that would cure or prevent diseases . What if you could make the genetic changes to eliminate diabetes or Alzheimer 's or reduce the risk of cancer or eliminate stroke ? Wouldn 't you want to make those changes in your genes ? If we look ahead , these kind of changes are going to be increasingly possible . The Human Genome Project started in 1990 , and it took 13 years . It cost 2.7 billion dollars .

➔ No wh-cleft

16

270652

Can you imagine something good coming out of tobacco ? These are all reality today , and [in] the future , will be evermore possible . Imagine then just two other little changes . You can change the cells in your body , but what if you could change the cells in your offspring ? What if you could change the sperm and the ova , or change the newly fertilized egg , and give your offspring a better chance at a healthier life -- eliminate the diabetes , eliminate the hemophilia , reduce the risk of cancer ? Who doesn 't want healthier children ? And then , that same analytic technology , that same engine of science that can produce the changes to prevent disease , will also enable us to adopt super-attributes , hyper-capacities -- that better memory . Why not have the quick wit of a Ken Jennings , especially if you can augment it with the next generation of the Watson machine ? Why not have the quick twitch muscle that will enable you to run faster and longer ?

➔ No wh-cleft

ID47:

271709

So nature already gives us really exquisite structures on the nanoscale . There are natural biological materials . The abalone shell here -- and if you fracture it , you can look at the fact that it 's nano-structured . There 's diatoms made out of SIO2 , and they 're magnetotactic bacteria that make small , single-domain magnets used for navigation . What all these have in common is these materials are structured at the nanoscale , and they have a DNA sequence that codes for a protein sequence that gives them the blueprint to be able to build these really wonderful structures . Now , going back to the abalone shell , the abalone makes this shell by having these proteins . These proteins are very negatively charged . And they can pull calcium out of the environment , put down a layer of calcium and then carbonate , calcium and carbonate . It has the chemical sequences of amino acids , which says , " This is how to build the structure .

ID48:

275634

This is a kind of a blowing straw . I just put this inside this . There 's a hole here , and I shut this . And this costs very little money to make -- great fun for children to do . What we do is make a very simple electric motor . If you have a battery , it costs five cents to make it . This is an old bicycle tube , which gives you a broad rubber band , two safety pins . This is a permanent magnet . Whenever current flows through the coil , this becomes an electromagnet .

ID49:

278374

And if the car and the driver were good enough , then you 'd win the race . Now today , if you want to win the race , actually you need also something like this -- something that monitors the car in real time , has a few thousand sensors collecting information from the car , transmitting this information into the system , and then processing it and using it in order to go back to the car with decisions and changing things in real time as information is collected . This is what , in engineering terms , you would call a real time control system . And basically , it 's a system made of two components -- a sensing and an actuating component . What is interesting today

is that real time control systems are starting to enter into our lives . Our cities , over the past few years , just have been blanketed with networks , electronics . They 're becoming like computers in open air . And , as computers in open air , they 're starting to respond in a different way to be able to be sensed and to be actuated . If we fix cities , actually it 's a big deal .

ID50:

281486

Using fruit and vegetable staining , create organic patterning . And using indigo , make it anti-microbial . And in fact , cotton would take up to 18 dips in indigo to achieve a color this dark . And because of the super-absorbency of this kind of cellulose , it just takes one , and a really short one at that . What I can 't yet do is make it water-resistant . So if I was to walk outside in the rain wearing this dress today , I would immediately start to absorb huge amounts of water . The dress would get really heavy , and eventually the seams would probably fall apart -- leaving me feeling rather naked . Possibly a good performance piece , but definitely not ideal for everyday wear . What I 'm looking for is a way to give the material the qualities that I need .

ID51:

281679

Make it hydrophobic . And while you 're at it , just form it around this 3D shape . " Bacterial cellulose is actually already being used for wound healing , and possibly in the future for biocompatible blood vessels , possibly even replacement bone tissue . But with synthetic biology , we can actually imagine engineering this bacterium to produce something that gives us the quality , quantity and shape of material that we desire . What excites me about using microbes is their efficiency . So we only grow what we need . There 's no waste . And in fact , we could make it from a waste stream -- so for example , a waste sugar stream from a food processing plant . Finally , at the end of use , we could biodegrade it naturally along with your vegetable peelings .

ID52:

282367

You might say , " Well , maybe things just smoothed themselves out . " But the conditions near the Big Bang are very , very different than the conditions of the air in this room . In particular , things were a lot denser . The gravitational pull of things was a lot stronger near the Big Bang . What you have to think about is we have a universe with a hundred billion galaxies , a hundred billion stars each . At early times , those hundred billion galaxies were squeezed into a region about this big -- literally -- at early times . And you have to imagine doing that squeezing without any imperfections , without any little spots where there were a few more atoms than somewhere else . Because if there had been , they would have collapsed under the gravitational pull into a huge black hole . Keeping the universe very , very smooth at early times is not easy ; it 's a delicate arrangement .

ID53:

288152

Darwin said , " I have never been an atheist in the same sense of denying the existence of a God . I think that generally an ' agnostic ' would be the most correct description of my state of mind . " He even became uncharacteristically tetchy with Edward Aveling . It was actually Edward Aveling . What happened was that Aveling 's mistress was Marx 's daughter , and when both Darwin and Marx were dead , Marx 's papers became muddled up with Aveling 's papers and a letter from Darwin saying , " My dear sir , thank you very much but I don 't want you to dedicate your book to me , " was mistakenly supposed to be addressed to Marx , and that gave rise to this whole myth , which you 've probably heard . It 's a sort of urban myth , that Marx tried to dedicate " Kapital " to Darwin . Anyway , it was Aveling , and when they met , Darwin challenged Aveling , " Why do you call yourselves atheists ? " " ' Agnostic , ' " retorted Aveling , " was simply ' atheist ' writ respectable , and ' atheist ' was simply ' agnostic ' writ aggressive . " Darwin complained , " But why should you be so aggressive ? "

ID54:

289566

And many scientists believe it 's the most serious issue facing mankind . It 's like the canary in the coalmine . If they disappear , so do we . It reminds us that we are a part of nature and we need to take care of it . What motivated me to film their behavior was something that I asked my scientific advisers : " What motivates the pollinators ? " Well , their answer was , " It 's all about risk and reward . " Like a wide-eyed kid , I 'd say , " Why is that ? " And they 'd say , " Well , because they want to survive . " I go , " Why ? "

17

290193

They 're more rare than the panda bear . I sat there on the river for two months without seeing one . I thought , my career 's over . I proposed this stupid story to National Geographic . What in the heck was I thinking ? So I had two months to sit there and figure out different ways of what I was going to do in my next life , after I was a photographer , because they were going to fire me . Because National Geographic is a magazine ; they remind us all the time : they publish pictures , not excuses . (Laughter) And after two months of sitting there -- one day , thinking that it was all over , this incredible big white male came down , right beside me , three feet away from me , and he went down and grabbed a fish and went off in the forest and ate it . And then I spent the entire day living my childhood dream of walking around with this bear through the forest .

➔ No wh-cleft

ID55:

293285

She would not stop trying to feed me penguins . And on the last day with this female where I thought I had pushed her too far , I got nervous because she came up to me , she rolled over on her back , and she did this deep , guttural jackhammer sound , this gokgokgokgok . And I thought , she 's about to bite . She 's about to let me know she 's too frustrated with me . What had happened was another seal had snuck in behind me , and she did that to threat display . She chased that big seal away , went and got its penguin and brought it to me . (Laughter) That wasn 't the only

seal I got in the water with . I got in the water with 30 other leopard seals , and I never once had a scary encounter . They are the most remarkable animals I 've ever worked with , and the same with polar bears .

ID56:

294476

You can imagine that then maybe you can use a pure protein to guide light , and so we 've made optical fibers . But silk is versatile and it goes beyond optics . And you can think of different formats . So for instance , if you 're afraid of going to the doctor and getting stuck with a needle , we do microneedle arrays . What you see there on the screen is a human hair superimposed on the needle that 's made of silk -- just to give you a sense of size . You can do bigger things . You can do gears and nuts and bolts -- that you can buy at Whole Foods . And the gears work in water as well . So you think of alternative mechanical parts .

ID57:

298225

And without skipping a beat at all , he said , " Well , they 'd be about the same . The only difference is a Susskind sandwich would have a lot more ham , " ham , as in bad actor . (Laughter) Well , I happened to have been very quick that day , and I said , " Yeah , but a lot less baloney . " (Laughter) The truth of the matter is that a Feynman sandwich had a load of ham , but absolutely no baloney . What Feynman hated worse than anything else was intellectual pretense – phoniness , false sophistication , jargon . I remember sometime during the ' 80s , the mid- ' 80s , Dick and I and Sidney Coleman would meet a couple of times up in San Francisco at some very rich guy 's house – up in San Francisco for dinner . And the last time the rich guy invited us , he also invited a couple of philosophers . These guys were philosophers of mind . Their specialty was the philosophy of consciousness .

18

302950

Let 's give an example of where we think this is going to go . Consider , for example , a condition like epilepsy , where the brain is overactive . Now if drugs fail in epileptic treatment , one of the strategies is to remove part of the brain . But that 's

obviously irreversible , and there could be side effects . What if we could just turn off that brain for a brief amount of time , until the seizure dies away , and cause the brain to be restored to its initial state – sort of like a dynamical system that ‘s being coaxed down into a stable state . So this animation just tries to explain this concept where we made these cells sensitive to being turned off with light , and we beam light in , and just for the time it takes to shut down a seizure , we ‘re hoping to be able to turn it off . And so we don ‘t have data to show you on this front , but we ‘re very excited about this . Now I want to close on one story , which we think is another possibility – which is that maybe these molecules , if you can do ultra-precise control , can be used in the brain itself to make a new kind of prosthetic , an optical prosthetic . I already told you that electrical stimulators are not uncommon .

➔ No wh-cleft

ID58:

312284

Being a visual artist , indeed , I am foremost interested to make art – to make art that transcends politics , religion , the question of feminism , and become an important , timeless , universal work of art . How to move you with your emotions , but also make your mind work . These are some of the images and the characters of the film . Now comes the green movement – the summer of 2009 , as my film is released – the uprising begins in the streets of Tehran . What is unbelievably ironic is the period that we tried to depict in the film , the cry for democracy and social justice , repeats itself now again in Tehran . The green movement significantly inspired the world . It brought a lot of attentio–to all those Iranians who stand for basic human rights and struggle for democracy . What was most significant for me was , once again , the presence of the women . They ‘re absolutely inspirational for me .

ID59:

312344

Now comes the green movement – the summer of 2009 , as my film is released – the uprising begins in the streets of Tehran . What is unbelievably ironic is the period that we tried to depict in the film , the cry for democracy and social justice ,

repeats itself now again in Tehran . The green movement significantly inspired the world . It brought a lot of attention to all those Iranians who stand for basic human rights and struggle for democracy . What was most significant for me was , once again , the presence of the women . They ‘re absolutely inspirational for me . If in the Islamic Revolution , the images of the woman portrayed were submissive and didn ‘t have a voice , now we saw a new idea of feminism in the streets of Tehran – women who were educated , forward thinking , non-traditional , sexually open , fearless and seriously feminist . These women and those young men united Iranians across the world , inside and outside . I then discovered why I take so much inspiration from Iranian women .

ID60:

315429

So in 2007 , our team won half a million dollars by placing third place in this competition . So about that time , the National Federation of the Blind , or NFB , challenged the research committee about who can develop a car that lets a blind person drive safely and independently . We decided to give it a try , because we thought , “ Hey , how hard could it be ? “ We have already an autonomous vehicle . What NFB wanted was not a vehicle that can drive a blind person around , but a vehicle where a blind person can make active decisions and drive . So we had to throw everything out the window and start from scratch . So to test this crazy idea , we developed a small dune buggy prototype vehicle to test the feasibility . It was an absolutely amazing experience . But the problem with this car was it was designed to only be driven in a very controlled environment , in a flat , closed-off parking lot – even the lanes defined by red traffic cones .

19

319451

And so we measured its motion . We found it was moving in really weird ways . And by giving it a gentle nudge , we were able to make it both vibrate and not vibrate at the same time – something that ‘s only allowed with quantum mechanics . So what I ‘m telling you here is something truly fantastic . What does it mean for one thing to be both vibrating and not vibrating at the same time ? So let ‘s think about the atoms . So in one case : all the trillions of atoms that make up that chunk

of metal are sitting still and at the same time those same atoms are moving up and down . Now it ‘s only at precise times when they align . The rest of the time they ‘re delocalized .

➔ No wh-cleft

20

321985

He thought it was impractical . Aristotle said , “ We should seek only so much precision in each subject as that subject allows . “ Aristotle thought ethics wasn ‘t a lot like math . He thought ethics was a matter of making decisions in the here-and-now using our best judgment to find the right path . How about this : What if in any situation you could just calculate , look at the choices , measure out which one ‘s better and know what to do ? That sound familiar ? That ‘s a utilitarian moral framework . ‘ohn Stuart Mill was a great advocate of this –nice guy besides – and only been dead 200 years . So basis of utilitarianism – I ‘m sure you ‘re familiar at least .

➔ No wh-cleft

21

328091

Many of these are a father who has to leave his family behind to go get a job -- sometimes a single mother who has to get enough money to even pay for food or clothing . Sadly , sometimes there are even children who are trying to get reunited with their parents that they haven 't seen , in some cases , for a decade . So what kind of an idea is it to think about building a brand new city in Honduras ? Or to build a dozen of these , or a hundred of these , around the world ? What kind of an idea is it to think about insisting that every family have a choice of several cities that are competing to attract new residents ? This is an idea worth spreading . And my friends from Honduras asked me to say thank you , TED . (Applause) So two couples each conceiving one baby .

➔ No wh-cleft

ID61:

328960

We 're making androgen , and we 're responding to androgens . The consequence is that somebody like me has actually had a brain exposed to more androgens than the woman born with testes who has androgen insensitivity syndrome . So sex is really complicated ; it 's not just that intersex people are in the middle of all the sex spectrum -- in some ways , they can be all over the place . Another example : a few years ago I got a call from a man who was 19 years old , who was born a boy , raised a boy , had a girlfriend , had sex with his girlfriend , had a life as a guy and had just found out that he had ovaries and a uterus inside . What he had was an extreme form of a condition called congenital adrenal hyperplasia . He had XX chromosomes , and in the womb , his adrenal glands were in such high gear that it created , essentially , a masculine hormonal environment . And as a consequence , his genitals were masculinzed , his brain was subject to the more typical masculine component of hormones . And he was born looking like a boy -- nobody suspected anything . Okay , so just one more quick example of a way you can have intersex .

ID62:

330014

So in order to tell you about that tension , I have to first admit to you that I 'm a huge fan of the Founding Fathers . I know they were racists , I know they were sexist , but they were great . I mean , they were so brave and so bold and so radical in what they did that I find myself watching that cheesy musical " 1776 " every few years , and it 's not because of the music , which is totally forgettable . The Founding Fathers were , for my point of view , the original anatomical activists , and this is why . What they rejected was an anatomical concept and replaced it with another one that was radical and beautiful and held us for 200 years . The monarchs of the old world didn 't have a concept of DNA , but they did have a concept of birthright . They had a concept of blue blood . They had the idea that the people who would be in political power should be in political power because of the blood being passed down from grandfather to father to son and so forth . The Founding Fathers rejected that idea , and they replaced it with a new anatomical concept , and that concept was all men are created equal .

ID63:

330371

So as part of that , they were coming to us with a concept that was about anatomical commonality . And in doing so , they were really setting up in a beautiful way the Civil Rights movement of the future . They didn 't think of it that way , but they did it for us , and it was great . So what happened years afterward ? What happened was women , for example , who wanted the right to vote , took the Founding Fathers ' concept of anatomical commonality being more important than anatomical difference and said , " The fact that we have a uterus and ovaries is not significant enough in terms of a difference to mean that we shouldn 't have the right to vote , the right to full citizenship , the right to own property , etc . , etc . " And women successfully argued that . Again , people of color appealing to a commonality of anatomy over a difference of anatomy , again , successfully . We see the same thing with the disability rights movement . The problem is , of course , that , as we begin to look at all that commonality , we have to begin to question why we maintain certain divisions .

ID64:

331337

It turns out to be really difficult . And part of me also wonders , what if content of character turns out to be something that 's scannable in the future -- able to be seen with an fMRI ? Do we really want to go there ? I 'm not sure where we go . What I do know is that it seems to be really important to think about the idea of the United States being in the lead of thinking about this issue of democracy . We 've done a really good job struggling with democracy , and I think we would do a good job in the future . We don 't have a situation that Iran has , for example , where a man who 's sexually attracted to other men is liable to be murdered , unless he 's willing to submit to a sex change , in which case he 's allowed to live . We don 't have that kind of situation . I 'm glad to say we don 't have the kind of situation with -- a surgeon I talked to a few years ago who had brought over a set of conjoined twins in order to separate them , partly to make a name for himself .

ID65:

334258

And whatever he makes somehow references nesting and space and personal property . This next project , this is by Jaochim Parisvega , and he 's interested in -- he believes art is everywhere waiting -- that it just needs a little bit of a push to happen . And he provides this push by harnessing natural forces , like in his series where he used rain to make paintings . This project is called " Love Nests . " What he did was to get wild birds to make his art for him . And this one 's called " Lovelock 's Nest . " Sylvia 's interested in art training . She 's a very serious Swiss artist . (Laughter) And she was thinking about her friends and family who work in chaos-ridden places and developing countries , and she was thinking , what can I make that would be of value to them , in case something bad happens and they have to buy their way across the border or pay off a gunman ?

22

335560

You know spiderwebs ? Sure , they are products of evolution . Beaver dams , yes . Hoover Dam , no . What do they think it is that prevents the products of human ingenuity from being themselves , fruits of the tree of life , and hence , in some sense , obeying evolutionary rules ? And yet people are interestingly resistant to the idea of applying evolutionary thinking to thinking -- to our thinking . And so I 'm going to talk a little bit about that , keeping in mind that we have a lot on the program here . So you 're out in the woods , or you 're out in the pasture , and you see this ant crawling up this blade of grass . It climbs up to the top , and it falls , and it climbs , and it falls , and it climbs -- trying to stay at the very top of the blade of grass .

➔ No wh-cleft

23

335718

So you 're out in the woods , or you 're out in the pasture , and you see this ant crawling up this blade of grass . It climbs up to the top , and it falls , and it climbs , and it falls , and it climbs -- trying to stay at the very top of the blade of grass . What

is this ant doing ? What is this in aid of ? What goals is this ant trying to achieve by climbing this blade of grass ? What 's in it for the ant ? And the answer is : nothing . There 's nothing in it for the ant . Is it just a fluke ?

➔ No wh-cleft

ID66:

337425

Work out the implications . There 's plenty of room for moral passion once we 've got the facts and can figure out the best thing to do . And , as with germs , the trick is not to try to annihilate them . You will never annihilate the germs . What you can do , however , is foster public health measures and the like that will encourage the evolution of avirulence . That will encourage the spread of relatively benign mutations of the most toxic varieties . That 's all the time I have , so thank you very much for your attention . I 'm here today to start a revolution . Now before you get up in arms , or you break into song , or you pick a favorite color , I want to define what I mean by revolution .

ID67:

340692

Unacceptable . Nepriemliv . (Laughter) So in summary , we need a drastic change in the way we think and behave , but we don 't need a workers ' revolution . We don 't need a workers ' revolution . What we need is a players ' uprising . What we need is a players ' uprising . What we need is a players ' uprising . Seriously , we need to band together . Today is the start of the uprising .

ID68:

340701

Nepriemliv . (Laughter) So in summary , we need a drastic change in the way we think and behave , but we don 't need a workers ' revolution . We don 't need a workers ' revolution . What we need is a players ' uprising . What we need is a players ' uprising . What we need is a players ' uprising . Seriously , we need to band together . Today is the start of the uprising . Because we can 't do it alone .

ID69:

340710

(Laughter) So in summary , we need a drastic change in the way we think and behave , but we don 't need a workers ' revolution . We don 't need a workers ' revolution . What we need is a players ' uprising . What we need is a players ' uprising . What we need is a players ' uprising . Seriously , we need to band together . Today is the start of the uprising . Because we can 't do it alone . Thank you .

24

345855

And of El Peje . And now of the Zetas and the traffickers and the criminals and the kidnappers ... Hold on ! Wait a minute ! What if none of these things is the problem ? The problem is not the things we feel victims of . The problem is that we play the role of victims . We need to open our eyes and see that we are not victims . If only we stopped feeling like victims , if we stopped acting as victims , our country would change so much !

ID70:

349276

Given the computer model , we can test the model by essentially quizzing it . So we could deliberately erase some symbols , and we can ask it to predict the missing symbols . Here are some examples . You may regard this as perhaps the most ancient game of Wheel of Fortune . What we found was that the computer was successful in 75 percent of the cases in predicting the correct symbol . In the rest of the cases , typically the second best guess or third best guess was the right answer . There 's also practical use for this particular procedure . There 's a lot of these texts that are damaged . Here 's an example of one such text .

25

349835

Who 'd have thought that deciphering could be a dangerous profession ? What does this result really show ? It shows that the Indus script shares an important property of language . So , as the old saying goes , if it looks like a linguistic script and it acts like a linguistic script , then perhaps we may have a linguistic script on our

hands . What other evidence is there that the script could actually encode language ? Well linguistic scripts can actually encode multiple languages . So for example , here 's the same sentence written in English and the same sentence written in Dutch using the same letters of the alphabet . If you don 't know Dutch and you only know English and I give you some words in Dutch , you 'll tell me that these words contain some very unusual patterns . Some things are not right , and you 'll say these words are probably not English words .

➔ No wh-cleft

ID71:
351914
She 's a veterinarian , so she 's seen lots of ultrasounds . I mean , she knows I 'm not a dog . But what we saw -- this is an MRI image . This is much sharper than an ultrasound would be . What we saw in that kidney was that big blob there . And there were actually two of these . I went home . Now I 've been Googling -- I 've been online since 1989 on CompuServe . My point is I went to a respected medical website , WebMD , because I know how to filter out junk .

ID72:
353624
So she pulled up the CD of the scan images , and look at all those hot spots . And she 's now actively engaged on her blog in looking for assistance in getting better care . See , that is an empowered patient -- no medical training . We are , you are , the most underused resource in health care . What she was able to do was because she had access to the raw data . Well at TED2009 , Tim Berners-Lee himself , inventor of the Web , gave a talk where he said the next big thing is not to have your browser go out and find other people 's articles about the data , but the raw data . And he got them chanting by the end of the talk , " Raw data now . Raw data now . " And I ask you , three words , please , to improve health care : let patients help .

ID73:

356599

Like for the hamburger , we just cut the damn barbecue . And so those coals that kept falling off the edge , we kept having to put them back up . But again , it only has to work for a thousandth of a second . The wok shot caught fire three times . What happens when you have your wok cut in half is the oil goes down into the fire and whoosh ! One of our cooks lost his eyebrows that way . But hey , they grow back . In addition to cutaways , we also explain physics . This is Fourier 's law of heat conduction .

ID74:

368487

Now , the viruses that we used to see in the 1980s and 1990s obviously are not a problem any more . So let me just show you a couple of examples of what they used to look like . What I 'm running here is a system that enables me to run age-old programs on a modern computer . So let me just mount some drives . What we have here is a list of old viruses . So let me just run some viruses on my computer . For example , let 's go with the Centipede virus first . And you can see at the top of the screen , there 's a centipede scrolling across your computer when you get infected by this one . You know that you 're infected because it actually shows up .

ID75:

368655

Yes , there 's a guy walking across your screen once you get infected . So it used to be fairly easy to know that you 're infected by a virus , when the viruses were written by hobbyists and teenagers . Today , they are no longer being written by hobbyists and teenagers . Today , viruses are a global problem . What we have here in the background is an example of our systems that we run in our labs , where we track virus infections worldwide . So we can actually see in real time that we 've just blocked viruses in Sweden and Taiwan and Russia and elsewhere . In fact , if I just connect back to our lab systems through the Web , we can see in real time just some kind of idea of how many viruses , how many new examples of malware we find every single day . Here 's the latest virus we 've found , in a file called Server.exe.

And we found it right over here three seconds ago -- the previous one , six seconds ago .

ID76:
369504
Now , criminals who weren 't capable of reaching us before can reach us . So how do you actually go around finding online criminals ? How do you actually track them down ? Let me give you an example . What we have here is one exploit file . Here , I 'm looking at the Hex dump of an image file , which contains an exploit . And that basically means , if you 're trying to view this image file on your Windows computer , it actually takes over your computer and runs code . Now , if you 'll take a look at this image file -- well there 's the image header , and there the actual code of the attack starts . And that code has been encrypted , so let 's decrypt it .

ID77:
370650
I 'm worried that we are running into problems because of online crime . Online crime is the one thing that might take these things away from us . (Laughter) I 've spent my life defending the Net , and I do feel that if we don 't fight online crime , we are running a risk of losing it all . We have to do this globally , and we have to do it right now . What we need is more global , international law enforcement work to find online criminal gangs -- these organized gangs that are making millions out of their attacks . That 's much more important than running anti-viruses or running firewalls . What actually matters is actually finding the people behind these attacks , and even more importantly , we have to find the people who are about to become part of this online world of crime , but haven 't yet done it . We have to find the people with the skills , but without the opportunities and give them the opportunities to use their skills for good . Thank you very much .

ID78:
370691
(Laughter) I 've spent my life defending the Net , and I do feel that if we don 't fight online crime , we are running a risk of losing it all . We have to do this globally , and we have to do it right now . What we need is more global , international law

enforcement work to find online criminal gangs -- these organized gangs that are making millions out of their attacks . That 's much more important than running anti-viruses or running firewalls . What actually matters is actually finding the people behind these attacks , and even more importantly , we have to find the people who are about to become part of this online world of crime , but haven 't yet done it . We have to find the people with the skills , but without the opportunities and give them the opportunities to use their skills for good . Thank you very much . (Applause) Embracing otherness .

ID79:

376622

This is what our whole economy is thrusting upon us , particularly illustrated in that left-hand corner : hockey sticks . This is a bunch of software companies -- and what it is is their revenue versus their age -- all zooming away , and everybody making millions and billions of dollars . Okay , so how do we understand this ? So let 's first talk about biology . What is plotted here is metabolic rate -- how much energy you need per day to stay alive -- versus your weight , your mass , for all of us bunch of organisms . And it 's plotted in this funny way by going up by factors of 10 , otherwise you couldn 't get everything on the graph . And what you see if you plot it in this slightly curious way is that everybody lies on the same line . Despite the fact that this is the most complex and diverse system in the universe , there 's an extraordinary simplicity being expressed by this . It 's particularly astonishing because each one of these organisms , each subsystem , each cell type , each gene , has evolved in its own unique environmental niche with its own unique history .

ID80:

378385

We could use sales , anything you like . There it is : after some little fluctuations at the beginning , when companies are innovating , they scale beautifully . And we 've looked at 23,000 companies in the United States , may I say . And I 'm only showing you a little bit of this . What is astonishing about companies is that they scale sublinearly like biology , indicating that they 're dominated , not by super-linear innovation and ideas ; they become dominated by economies of scale . In that interpretation , by bureaucracy and administration , and they do it beautifully , may

I say . So if you tell me the size of some company , some small company , I could have predicted the size of Walmart . If it has this sublinear scaling , the theory says we should have sigmoidal growth . There 's Walmart .

ID81:
379393
Many sociologists like Veblen and Wolfe would argue that the reason why we take origins so seriously is because we 're snobs , because we 're focused on status . Among other things , if you want to show off how rich you are , how powerful you are , it 's always better to own an original than a forgery because there 's always going to be fewer originals than forgeries . I don 't doubt that that plays some role , but what I want to convince you of today is that there 's something else going on . I want to convince you that humans are , to some extent , natural born essentialists . What I mean by this is we don 't just respond to things as we see them , or feel them , or hear them . Rather , our response is conditioned on our beliefs , about what they really are , what they came from , what they 're made of , what their hidden nature is . I want to suggest that this is true , not just for how we think about things , but how we react to things . So I want to suggest that pleasure is deep -- and that this isn 't true just for higher level pleasures like art , but even the most seemingly simple pleasures are affected by our beliefs about hidden essences . So take food .

ID82:
381604
(Laughter) For a dollar 99 , you can listen to that silence , which is different than other forms of silence . (Laughter) Now I 've been talking so far about pleasure , but what I want to suggest is that everything I 've said applies as well to pain . And how you think about what you 're experiencing , your beliefs about the essence of it , affect how it hurts . One lovely experiment was done by Kurt Gray and Dan Wegner . What they did was they hooked up Harvard undergraduates to an electric shock machine . And they gave them a series of painful electric shocks . So it was a series of five painful shocks . Half of them are told that they 're being given the shocks by somebody in another room , but the person in the other room doesn 't know they 're giving them shocks . There 's no malevolence , they 're just pressing a button .

ID83:

383405

The next thing I want to talk about is school feeding . Eighty percent of the people in the world have no food safety net . When disaster strikes -- the economy gets blown , people lose a job , floods , war , conflict , bad governance , all of those things -- there is nothing to fall back on . And usually the institutions -- churches , temples , other things -- do not have the resources to provide a safety net . What we have found working with the World Bank is that the poor man 's safety net , the best investment , is school feeding . And if you fill the cup with local agriculture from small farmers , you have a transformative effect . Many kids in the world can 't go to school because they have to go beg and find a meal . But when that food is there , it 's transformative . It costs less than 25 cents a day to change a kid 's life .

ID84:

384847

Because food is one issue that cannot be solved person by person , nation by nation . We have to stand together . And we 're seeing nations in Africa . WFP 's been able to leave 30 nations because they have transformed the face of hunger in their nations . What I would like to offer here is a challenge . I believe we 're living at a time in human history where it 's just simply unacceptable that children wake up and don 't know where to find a cup of food . Not only that , transforming hunger is an opportunity , but I think we have to change our mindsets . I am so honored to be here with some of the world 's top innovators and thinkers . And I would like you to join with all of humanity to draw a line in the sand and say , " No more .

ID85:

386937

So that when Louis Leakey , my mentor , heard this news , he said , " Ah , we must now redefine ' man , ' redefine ' tool , ' or accept chimpanzees as humans . " (Laughter) We now know that at Gombe alone , there are nine different ways in which chimpanzees use different objects for different purposes . Moreover , we know that in different parts of Africa , wherever chimps have been studied , there are completely different tool-using behaviors . And because it seems that these patterns are passed from one generation to the next , through observation , imitation

and practice -- that is a definition of human culture . What we find is that over these 40-odd years that I and others have been studying chimpanzees and the other great apes , and , as I say , other mammals with complex brains and social systems , we have found that after all , there isn 't a sharp line dividing humans from the rest of the animal kingdom . It 's a very wuzzy line . It 's getting wuzzier all the time as we find animals doing things that we , in our arrogance , used to think was just human . The chimps -- there 's no time to discuss their fascinating lives -- but they have this long childhood , five years of suckling and sleeping with the mother , and then another three , four or five years of emotional dependence on her , even when the next child is born . The importance of learning in that time , when behavior is flexible -- and there 's an awful lot to learn in chimpanzee society .

ID86:
391759
(Applause) By the end of this year , there 'll be nearly a billion people on this planet that actively use social networking sites . The one thing that all of them have in common is that they 're going to die . While that might be a somewhat morbid thought , I think it has some really profound implications that are worth exploring . What first got me thinking about this was a blog post authored earlier this year by Derek K. Miller , who was a science and technology journalist who died of cancer . And what Miller did was have his family and friends write a post that went out shortly after he died . Here 's what he wrote in starting that out . He said , " Here it is . I 'm dead , and this is my last post to my blog .

ID87:
393598
Let 's go to the closest neighbor to the visible light spectrum -- go to remote controls . You all know remote controls have an infrared LED -- basically you switch on the LED , and if it 's off , you switch it off . And it creates a simple , low-speed data stream in 10,000 bits per second , 20,000 bits per second . Not usable for a YouTube video . What we have done is we have developed a technology with which we can furthermore replace the remote control of our light bulb . We transmit with our technology , not only a single data stream , we transmit thousands of data streams in parallel , at even higher speeds . And the technology we have developed -- it 's

called SIM OFDM . And it 's spacial modulation -- these are the only technical terms , I 'm not going into details -- but this is how we enabled that light source to transmit data . You will say , " Okay , this is nice -- a slide created in 10 minutes."

ID88:

398315

So many of you will be familiar with the idea that there 's this gene called FOXP2 that seems to be implicated in some ways in the fine motor control that 's associated with language . The reason why I don 't believe that tells us that the Neanderthals had language is -- here 's a simple analogy : Ferraris are cars that have engines . My car has an engine , but it 's not a Ferrari . Now the simple answer then is that genes alone don 't , all by themselves , determine the outcome of very complicated things like language . What we know about this FOXP2 and Neanderthals is that they may have had fine motor control of their mouths -- who knows . But that doesn 't tell us they necessarily had language . MR : Thank you very much indeed . (Applause) Humans in the developed world spend more than 90 percent of their lives indoors , where they breathe in and come into contact with trillions of life forms invisible to the naked eye : microorganisms .

ID89:

401232

It was the continuing devastating war in the Congo and the indifference of the world . It was the Congolese women who are now rising up . It was my mother leaving , just at the moment that I was being born . It was the realization that I had come very close to dying -- in the same way that the Earth , our mother , is barely holding on , in the same way that 75 percent of the planet are hardly scraping by , in the same way that there is a recipe for survival . What I learned is it has to do with attention and resources that everybody deserves . It was advocating friends and a doting sister . It was wise doctors and advanced medicine and surgeons who knew what to do with their hands . It was underpaid and really loving nurses . It was magic healers and aromatic oils .

ID90:

403015

And it is possible , of course , to increase the density of the communities around us . So we 're talking about things like infill development : really sharp little changes to where we have buildings , where we 're developing . Urban retrofitting : creating different sorts of spaces and uses out of places that are already there . Increasingly , we 're realizing that we don 't even need to densify an entire city . What we need instead is an average density that rises to a level where we don 't drive as much and so on . And that can be done by raising the density in very specific spots a whole lot . So you can think of it as tent poles that actually raise the density of the entire city . And we find that when we do that , we can , in fact , have a few places that are really hyper-dense within a wider fabric of places that are perhaps a little more comfortable and achieve the same results . We have the manufacture of the car , the disposal of the car , all of the parking and freeways and so on .

26

411676

" Fantastic ! " I thought . " Let 's throw him out . Who would ever include a drunken guy in a sample ? " But a couple of days later , we thought about it with my students , and we said , " What would have happened if this drunken guy was not in that condition ? What would have happened if he was in the other group ? What a smart guy who is performing this low , " because he would have pulled the mean of the group lower , giving us even stronger statistical results than we could . So we decided not to throw the guy out and to rerun the experiment . But you know , these stories , and lots of other experiments that we 've done on conflicts of interest , basically kind of bring two points to the foreground for me .

➔ No wh-cleft

27

411690

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➔ No wh-cleft

28

411702

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➔ No wh-cleft

ID91:

412156

The positive perspective , I think , of all of this is that , if we do understand when we go wrong , if we understand the deep mechanisms of why we fail and where we fail , we can actually hope to fix things . And that , I think , is the hope . Thank you very much . (Applause) What I want to talk to you about is what we can learn from studying the genomes of living people and extinct humans . But before doing that , I just briefly want to remind you about what you already know : that our genomes , our genetic material , are stored in almost all cells in our bodies in chromosomes in

the form of DNA , which is this famous double-helical molecule . And the genetic information is contained in the form of a sequence of four bases abbreviated with the letters A , T , C and G. And the information is there twice -- one on each strand -- which is important , because when new cells are formed , these strands come apart , new strands are synthesized with the old ones as templates in an almost perfect process . But nothing , of course , in nature is totally perfect , so sometimes an error is made and a wrong letter is built in .

ID92:
415513
There is no need to use violence . Chanting : We can do it ! We can do it ! We can do it ! What was even more surprising was the fact that Budrus was successful . The residents , after 10 months of peaceful resistance , convinced the Israeli government to move the route of the barrier off their lands and to the green line , which is the internationally recognized boundary between Israel and the Palestinian Territories . The resistance in Budrus has since spread to villages across the West Bank and to Palestinian neighborhoods in Jerusalem . Yet the media remains mostly silent on these stories . This silence carries profound consequences for the likelihood that nonviolence can grow , or even survive , in Palestine .

ID93:
419947
So were the followers of Hippocrates . The Hippocratic manuscripts also -- repeatedly , according to recent studies -- show how important it is not to do harm . More recently , Harvey Cushing , who really developed neurosurgery as we know it , who changed it from a field of medicine that had a majority of deaths resulting from surgery to one in which there was a hopeful outlook , he was very conscious that he was not always going to do the right thing . But he did his best , and he kept meticulous records that let him transform that branch of medicine . What we find is , no longer simple tools , but systems . We find more and more complex arrangements of machines that make it harder and harder to diagnose what 's going on . Thomas Edison would have been very , very comfortable in the atmosphere of a software firm today . And these hackers had a word for those mysterious bugs in telegraph systems that they called bugs . That was the origin of the word " bug . "

ID94:

420626

Well , technology to the rescue . So chemists got to work , and they developed a bactericide that became widely used in those systems . But something else happened in the early 1980s , and that was that there was a mysterious epidemic of failures of tape drives all over the United States . And IBM , which made them , just didn 't know what to do . What happened was the bactericide was formulated with minute traces of tin . And these tin particles were deposited on the tape heads and were crashing the tape heads . So they reformulated the bactericide . But what 's interesting to me is that this was the first case of a mechanical device suffering , at least indirectly , from a human disease . So it shows that we 're really all in this together .

29

421673

And evolution , although it was established well over 100 years ago by Charles Darwin and a vast number of other people , evolution still is a little bit intangible . And when I talk about Darwinian evolution , I mean one thing and one thing only , and that is survival of the fittest . And so forget about evolution in a kind of metaphysical way . Think about evolution in terms of offspring competing , and some winning . So bearing that in mind , as a chemist , I wanted to ask myself the question frustrated by biology : What is the minimal unit of matter that can undergo Darwinian evolution ? And this seems quite a profound question . And as a chemist , we 're not used to profound questions every day . So when I thought about it , then suddenly I realized that biology gave us the answer . And in fact , the smallest unit of matter that can evolve independently is , in fact , a single cell -- a bacteria .

➔ No wh-cleft

ID95:

423725

So that means we could then start to develop different platforms for exploring this evolution . So you imagine , if we 're able to create a self-sustaining artificial life form , not only will this tell us about the origin of life -- that it 's possible that the universe doesn 't need carbon to be alive ; it can use anything -- we can then take [

it] one step further and develop new technologies , because we can then use software control for evolution to code in . So imagine we make a little cell . We want to put it out in the environment , and we want it to be powered by the Sun . What we do is we evolve it in a box with a light on . And we don 't use design anymore . We find what works . We should take our inspiration from biology . Biology doesn 't care about the design unless it works .

ID96:
432816
So unfortunately , this doesn 't help this woman , but it does have severe , profound if you will , implications to her family . I mean , if they have the same mutation , and they get this genetic test , and they understand it , then they can go and get regular screens , and they can catch cancer early and potentially live a significantly longer life . Let me introduce you now to the Beery twins , diagnosed with cerebral palsy at the age of two . Their mom is a very brave woman who didn 't believe that the symptoms weren 't matching up , and through some heroic efforts and a lot of Internet searching , she was able to convince the medical community that , in fact , they had something else . What they had was dopa-responsive dystonia . And so they were given L-Dopa , and their symptoms did improve , but they weren 't totally asymptomatic . Significant problems remained . Turns out the gentleman in this picture is a guy named Joe Beery , who was lucky enough to be the CIO of a company called Life Technologies . They 're one of the two companies that makes these massive whole genome sequencing tools .

ID97:
438918
Now the problem is there 's an X-axis for that , which is the practical axis . This is very , very low . (Applause) Now people tend to use an alternative approach , which is to take a few sources and read them very carefully . This is extremely practical , but not so awesome . What you really want to do is to get to the awesome yet practical part of this space . So it turns out there was a company across the river called Google who had started a digitization project a few years back that might just enable this approach . They have digitized millions of books . So what that means

is , one could use computational methods to read all of the books in a click of a button . That 's very practical and extremely awesome .

ID98:
440358
If the suppression index is very , very , very small , then you very well might be being suppressed . If it 's very large , maybe you 're benefiting from propaganda . JM : Now you can actually look at the distribution of suppression indexes over whole populations . So for instance , here -- this suppression index is for 5,000 people picked in English books where there 's no known suppression -- it would be like this , basically tightly centered on one . What you expect is basically what you observe . This is distribution as seen in Germany -- very different , it 's shifted to the left . People talked about it twice less as it should have been . But much more importantly , the distribution is much wider . There are many people who end up on the far left on this distribution who are talked about 10 times fewer than they should have been .

ID99:
442983
I took these back to the lab . I extracted DNA from them . And when I looked at the sequence of the DNA , and compared the sequence of Jonas ' tumor to that of the rest of his body , I discovered that they had a completely different genetic profile . In fact , Jonas and his tumor were as different from each other as you and the person sitting next to you . What this told us was that Jonas ' tumor did not arise from cells of his own body . In fact , more genetic profiling told us that this tumor in Jonas actually probably first arose from the cells of a female Tasmanian devil -- and Jonas was clearly a male . So how come a tumor that arose from the cells of another individual is growing on Jonas ' face ? Well the next breakthrough came from studying hundreds of Tasmanian devil cancers from all around Tasmania . We found that all of these cancers shared the same DNA .

ID100:

448151

I will never abandon you . I will be with you through the end . " Thank you very much . (Applause) What I want to talk about today is one idea . It 's an idea for a new kind of school , which turns on its head much of our conventional thinking about what schools are for and how they work . And it might just be coming to a neighborhood near you soon . Where it comes from is an organization called the Young Foundation , which , over many decades , has come up with many innovations in education , like the Open University and things like Extended Schools , Schools for Social Entrepreneurs , Summer Universities and the School of Everything . One was large numbers of bored teenagers who just didn 't like school , couldn 't see any relationship between what they learned in school and future jobs .

ID101:

452143

We know from three different studies on three different types of pain that a saltwater injection is a more effective treatment for pain than taking a sugar pill , taking a dummy pill that has no medicine in it -- not because the injection or the pills do anything physically to the body , but because an injection feels like a much more dramatic intervention . So we know that our beliefs and expectations can be manipulated , which is why we do trials where we control against a placebo -- where one half of the people get the real treatment and the other half get placebo . But that 's not enough . What I 've just shown you are examples of the very simple and straightforward ways that journalists and food supplement pill peddlers and naturopaths can distort evidence for their own purposes . What I find really fascinating is that the pharmaceutical industry uses exactly the same kinds of tricks and devices , but slightly more sophisticated versions of them , in order to distort the evidence that they give to doctors and patients , and which we use to make vitally important decisions . So firstly , trials against placebo : everybody thinks they know that a trial should be a comparison of your new drug against placebo . But actually in a lot of situations that 's wrong . Because often we already have a very good treatment that is currently available , so we don 't want to know that your

alternative new treatment is better than nothing . We want to know that it 's better than the best currently available treatment that we have .

ID102:
459602
Basically , there 's a major demographic event going on . And it may be that passing the 50 percent urban point is an economic tipping point . So the world now is a map of connectivity . It used to be that Paris and London and New York were the largest cities . What we have now is the end of the rise of the West . That 's over . The aggregate numbers are overwhelming . So what 's really going on ? Well , villages of the world are emptying out .

ID103:
470192
And these were surfaced by researcher David Matsumoto in California . And I think they 're an excellent example of what the truth looks like . This mother , Diane Downs , shot her kids at close range , drove them to the hospital while they bled all over the car , claimed a scraggy-haired stranger did it . And you 'll see when you see the video , she can 't even pretend to be an agonizing mother . What you want to look for here is an incredible discrepancy between horrific events that she describes and her very , very cool demeanor . And if you look closely , you 'll see duping delight throughout this video . (Video) Diane Downs : At night when I close my eyes , I can see Christie reaching her hand out to me while I 'm driving , and the blood just kept coming out of her mouth . And that -- maybe it 'll fade too with time -- but I don 't think so . That bothers me the most .

ID104:
472411
You went to the most expensive school and college in India , and you want to dig wells for five years ? " She didn 't speak to me for a very long time , because she thought I 'd let my family down . But then , I was exposed to the most extraordinary knowledge and skills that very poor people have , which are never brought into the mainstream -- which is never identified , respected , applied on a large scale . And I thought I 'd start a Barefoot College -- college only for the poor . What the poor

thought was important would be reflected in the college . I went to this village for the first time . Elders came to me and said , " Are you running from the police ? " Why are you here ? The education system in India makes you look at Paris and New Delhi and Zurich ; what are you doing in this village ?

ID105:
476821
And together , these variables construct a score . I use these scores to collaborate with musicians . This is the 1913 Trio performing one of my pieces at the Milwaukee Art Museum . Meanwhile , I use these scores as blueprints to translate into sculptural forms like this , that function still in the sense of being a three-dimensional weather visualization , but now they 're embedding the visual matrix of the musical score , so it can actually be read as a musical score . What I love about this work is that it challenges our assumptions of what kind of visual vocabulary belongs in the world of art , versus science . This piece here is read very differently depending on where you place it . You place it in an art museum , it becomes a sculpture . You place it in a science museum , it becomes a three-dimensional visualization of data . You place it in a music hall , it all of a sudden becomes a musical score .

ID106:
484221
And everyone in the lab , especially following the interaction with these patients , feels quite compelled to deliver a drug substance based on this molecule . It 's here where I have to say that we could use your help and your insights , your collaborative participation . Unlike a drug company , we don 't have a pipeline that we can deposit these molecules into . We don 't have a team of salespeople and marketeers that can tell us how to position this drug against the other . What we do have is the flexibility of an academic center to work with competent , motivated , enthusiastic , hopefully well-funded people to carry these molecules forward into the clinic while preserving our ability to share the prototype drug worldwide . This molecule will soon leave our benches and go into a small startup company called Tensha Therapeutics . And really this is the fourth of these molecules to kind of graduate from our little pipeline of drug discovery , two of which -- a topical drug

for lymphoma of the skin , an oral substance for the treatment of multiple myeloma -- will actually come to the bedside for first clinical trial in July of this year . For us , a major and exciting milestone . I want to leave you with just two ideas .

30

491631

So one Sunday afternoon , a man walks into my cashier 's booth with this beautiful jewelry box . Opens it up and there 's a pearl necklace inside . And he said , " Hey , I was in the men 's room . I just found this . What do you think we should do with it ? " " I don 't know , put it in the lost and found . " " Well this is very valuable . I said , " Yea . " So we 're trying to decide what to do with this , and the phone rings .

31

491830

Get yourself a fat reward . Life 's good . " He said , " I can 't do it . I have this job interview in Galena in 15 minutes , and I need this job , I 've got to go . " Again he asked me , " What do you think we should do ? " I 'm in high school . I have no idea . So I said , " I 'll hold it for you . " He said , " You know , you 've been so nice , let 's split the reward . "

ID107:

493729

But the last article in that issue was written by Francis Crick of DNA fame . Today is , I think , the 50th anniversary of the discovery of DNA . And he wrote a story basically saying , well , this is all well and good , but you know what , we don 't know diddley squat about brains and no one has a clue how these things work , so don 't believe what anyone tells you . This is a quote from that article . He said , " What is conspicuously lacking , " he 's a very proper British gentleman so , " What is conspicuously lacking is a broad framework of ideas in which to interpret these different approaches . " I thought the word framework was great . He didn 't say we didn 't even have a theory . He says , we don 't even know how to begin to think about it -- we don 't even have a framework . We are in the pre-paradigm days , if you want to use Thomas Kuhn .

ID108:

495925

All the mountains and the continents are kind of floating around on top of the Earth , you know ? It 's like , it doesn 't make any sense . So what is the intuitive , but incorrect assumption , that 's kept us from understanding brains ? The intuitive but obvious thing is that somehow intelligence is defined by behavior , that we are intelligent because of the way that we do things and the way we behave intelligently , and I 'm going to tell you that 's wrong . What it is is intelligence is defined by prediction . And I 'm going to work you through this in a few slides here , give you an example of what this means . Here 's a system . Engineers like to look at systems like this . Scientists like to look at systems like this .